TUH meta features — what do we know about the patients and recordings?

📄 **PDF version**

Part I (Sections 1–7) covers TUAB. **Part II** (Sections 8–15) repeats the analysis on the TUEG brain-age cohort `all_both_v3` (29k recordings), where NEDC's clinical sheet adds diagnoses, medications and the report's reason for an abnormal read.

The fine-tuning task only ever sees **EEG windows + a binary normal/abnormal label**. This notebook takes stock of everything *else* we know about each recording, and asks whether any of it is useful for training or prediction.

Sources of meta information (all read-only):

Source	What it gives	How
Directory layout	normal/abnormal label, official train/eval split	path
EDF patient field	**age**, **sex**, subject id	`labram/data/tuh_metadata.py`
EDF recording field	**session number** (patient's n-th TUH EEG), **year**, equipment tag	same
EDF signal headers	sampling rate, amplifier range, channel inventory (EKG, EMG, EOG, photic, **sphenoidal electrodes**)	`labram/data/tuh_meta_features.py`
Auxiliary channels	**heart rate / HRV** from the EKG lead, **photic stimulation** performed, device-computed **burst-suppression trends**	same
The EEG itself	**signal quality** (clipping, flat/noisy electrodes, artifacts, mains noise) and **qEEG** summaries (slowing, posterior dominant rhythm, asymmetry, sharp transients)	same
Processed windows	which recordings made it into `processed/` and which split	`processed_npy/manifest.json`

The per-recording table is built once by `python dataset_maker/make_TUAB_meta.py --root <edf dir>` (≈25 min, 8 cores) and cached as `processed/meta_features.csv`; this notebook only loads and analyses it.

```
2993 recordings, 2329 subjects, 63 columns
extraction errors: 0
```

tuh_split	processed_split / label	test	train	val	— not processed —	All
eval	**abnormal**	126	0	0	0	126
	normal	150	0	0	0	150
train	**abnormal**	0	1052	291	3	1346
	normal	0	1090	281	0	1371
All		276	2142	572	3	2993

1. Inventory — which meta features exist, and how complete are they?

Coverage = share of the 2,993 recordings where the feature has a value. A feature can be missing because the channel was not recorded (EMG, photic), because the device did not compute it (burst-suppression trends), or because it is a sentinel (age 999 = 90+).

Columns of the inventory table (`labram/eval/meta_analysis.py::feature_inventory`):

- **type**: *categorical* for 0/1 flags and discrete device settings (sampling rate, amplifier range); *quantitative* otherwise.
- **n_categories** / **categories** (categorical only): the number of distinct values, and the 5 most frequent ones with their share of the recordings that have a value. For 0/1 flags, 1 = yes.
- **active_range** (quantitative only): the 1st to 99th percentile of the values, $[Q_{0.01}(x),\ Q_{0.99}(x)]$, so a few outliers do not set the range.
- **meaning**: what a *history*, *acquisition*, *physiology* or *qEEG* feature measures, with its unit.

group	feature	source	type	coverage_%	n_categories	categories	active_range	meaning
demographics	age	EDF patient field	quantitative	99.5			17 – 86	
demographics	sex_male	EDF patient field	categorical	100.0	2	0 (53%), 1 (47%)		
history	session_number	EDF recording field	quantitative	100.0			1 – 13	Which EEG session of the patient this is: n-th TUH EEG session, in date order over the whole TUH archive (s003 -> 3). NOT the patient's total number of sessions
history	year	EDF recording field	quantitative	100.0			2000 – 2013	Recording year (the date is anonymised to 01-JAN-YYYY); 1899 and 2000 are placeholders
acquisition	duration_min	EDF signal headers / photic channel	quantitative	100.0			16.8 – 51.1	Length of the recording [min]
acquisition	sfreq	EDF signal headers / photic channel	categorical	100.0	3	250 (93%), 256 (6%), 512 (1%)		Native sampling rate of the EEG channels [Hz]; set by the amplifier / acquisition software
acquisition	physical_max_uv	EDF signal headers / photic channel	categorical	100.0	2	5000 (93%), 5482 (7%)		Amplifier input range from the EDF header [±µV]; identifies the amplifier family
acquisition	n_signals	EDF signal headers / photic channel	quantitative	100.0			27 – 36	Channels stored in the EDF (EEG + auxiliary); a montage / amplifier fingerprint
acquisition	has_emg	EDF signal headers / photic channel	categorical	100.0	2	1 (61%), 0 (39%)		An EMG (muscle) channel was recorded
acquisition	has_eog	EDF signal headers / photic channel	categorical	100.0	2	1 (93%), 0 (7%)		Both eye-movement channels (LOC, ROC) were recorded
acquisition	has_photic_channel	EDF signal headers / photic channel	categorical	100.0	2	1 (95%), 0 (5%)		The photic-stimulator trigger channel is present
acquisition	has_sphenoidal	EDF signal headers / photic channel	categorical	100.0	2	0 (96%), 1 (4%)		Sphenoidal electrodes (SP1/SP2), placed for temporal-lobe epilepsy work-ups
acquisition	has_numbered_extra	EDF signal headers / photic channel	categorical	100.0	2	1 (59%), 0 (41%)		Spare numbered amplifier inputs are stored (e.g. '26'...'32')
acquisition	photic_performed	EDF signal headers / photic channel	categorical	100.0	2	1 (72%), 0 (28%)		Photic stimulation was actually delivered (stimulator channel not constant)
acquisition	photic_stim_sec	EDF signal headers / photic channel	quantitative	94.8			0 – 125	Seconds during which the stimulator flashed [s]
quality	q_clip_frac	EEG signal	quantitative	100.0			0 – 0.000452	
quality	q_flat_epoch_frac	EEG signal	quantitative	100.0			0 – 0.022	
quality	q_high_amp_epoch_frac	EEG signal	quantitative	100.0			0 – 0.487	
quality	q_bad_epoch_frac	EEG signal	quantitative	100.0			0 – 0.726	
quality	q_n_flat_channels	EEG signal	quantitative	100.0			0 – 0	
quality	q_n_noisy_channels	EEG signal	quantitative	100.0			0 – 6	
quality	q_line60_log_ratio	EEG signal	quantitative	100.0			-3.41 – 0.248	
physiology	ekg_hr_bpm	EKG lead / device trend channels	quantitative	100.0			49 – 127	Heart rate from R-peaks of the EKG lead (median R-R interval) [beats/min]
physiology	ekg_rr_std_ms	EKG lead / device trend channels	quantitative	100.0			7.36 – 288	Standard deviation of the R-R intervals, a heart-rate-variability measure [ms]
physiology	trend_suppr_mean	EKG lead / device trend channels	quantitative	60.3			0.153 – 98.8	Mean of the device's SUPPR trend [%]; semantics unclear (higher in normals), not a clinical burst-suppression ratio
physiology	trend_suppr_gt50_frac	EKG lead / device trend channels	quantitative	60.3			0 – 0.997	Share of seconds with the SUPPR trend above 50%

group	feature	source	type	cover-age_%	n_categories	categories	active_range	meaning
physiol-ogy	trend_bursts_mean	EKG lead / device trend channels	quan-tita-tive	60.3			0.0181 – 1.87	Mean of the device's BURSTS trend [bursts/min]
qeeg	qeeg_rel_delta	EEG signal	quan-tita-tive	100.0			0.11 – 0.899	Delta (1-4 Hz) power / total 1-30 Hz power; high = slowing
qeeg	qeeg_rel_theta	EEG signal	quan-tita-tive	100.0			0.0475 – 0.405	Theta (4-8 Hz) power / total 1-30 Hz power; high = slowing / drowsiness
qeeg	qeeg_rel_alpha	EEG signal	quan-tita-tive	100.0			0.0147 – 0.607	Alpha (8-13 Hz) power / total 1-30 Hz power; the normal awake rhythm
qeeg	qeeg_rel_beta	EEG signal	quan-tita-tive	100.0			0.00926 – 0.51	Beta (13-30 Hz) power / total 1-30 Hz power; fast activity
qeeg	qeeg_log_total_power	EEG signal	quan-tita-tive	100.0			1.07 – 3.01	log10 of the 1-30 Hz power [μV²]; low = low-voltage EEG
qeeg	qeeg_theta_alpha_ratio	EEG signal	quan-tita-tive	100.0			-0.863 – 0.85	log10(relative theta / relative alpha); rises as the background slows
qeeg	qeeg_slowing_ratio	EEG signal	quan-tita-tive	100.0			-0.566 – 1.58	log10((delta + theta) / (alpha + beta)); dif-fuse-slowing index
qeeg	qeeg_pdr_freq	EEG signal	quan-tita-tive	100.0			5 – 14	Posterior dominant rhythm: frequency of the largest 5-14 Hz peak above the 1/f fit over O1/O2/P3/P4 [Hz]; < 8 Hz in adults = slow
qeeg	qeeg_pdr_prominence	EEG signal	quan-tita-tive	100.0			0.0102 – 1.62	Height of that peak above the 1/f fit [log10 power]; ~0 = no distinct PDR
qeeg	qeeg_aperiodic_slope	EEG signal	quan-tita-tive	100.0			-3.12 – -0.25	Slope of log power vs log frequency, 2-30 Hz, posterior channels (the 1/f exponent); more negative = steeper
qeeg	qeeg_max_asymmetry	EEG signal	quan-tita-tive	100.0			0.0556 – 0.87	Largest |log10(left / right)| total power over 8 homologous pairs; hemispheric asymmetry
qeeg	qeeg_max_delta_asymmetry	EEG signal	quan-tita-tive	100.0			0.0563 – 0.925	Same for delta power only; focal (one-sided) slowing
qeeg	qeeg_median_kurtosis	EEG signal	quan-tita-tive	100.0			0.291 – 4187	Median over channels of the excess kurto-sis of the signal; spikes, sharp waves (and eye blinks) make it heavy-tailed

	recordings	%
Age missing / sentinel (999 = 90+, 0)	15	0.501
Sex missing	0	0.000
No EKG channel	0	0.000
EKG present but no usable heart rate	0	0.000
No EMG channel	1179	39.392
No photic channel	157	5.246
Photic channel but no stimulation	680	22.720
Burst-suppression trend not computed	1187	39.659
EDF not in processed windows	3	0.100

What is *not* in the data at all — worth stating explicitly, because it bounds every question below:

Wanted	Available?
Reason the EEG is abnormal (slowing, epileptiform dis-charges, seizure, …)	❌ TUAB labels are binary. TUH stopped distributing the EEG reports, which held the findings.
Diagnosis, medications, indication for the EEG, ICU status	❌ (only in the withdrawn reports)
Race/ethnicity, handedness, height/weight	❌
Vigilance state (awake/drowsy/asleep), eyes open/closed events	❌ not annotated in TUAB (only indirectly: photic stimulation, PDR prominence)
Exact recording date / time of day	❌ anonymised to 01-JAN-YYYY 00:00 — only the **year** survives
Per-recording label confidence	⚠️ corpus-level only: the AAREADME reports 27 train files (1%) where the annotator disagreed with the report, but not *which*

2. Demographics — age, sex, and patient history

Session number = *which* EEG session of the patient a recording belongs to: the recording's session tag counts the patient's TUH EEG sessions in date order over the whole TUH archive (`s001` = first EEG at Temple, `s003` = third). It is **not** the patient's total number of sessions, and not a count within TUAB: for 23% of TUAB patients the earliest TUAB recording is already

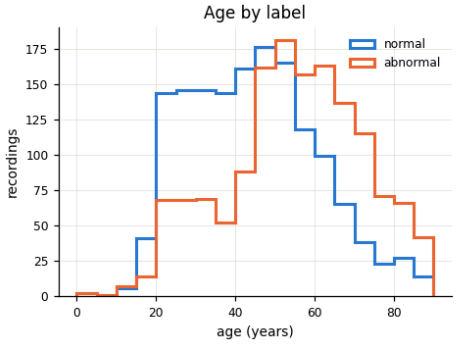
session 2 or later. A high session number therefore means a patient who had already been recorded many times — chronic neurological follow-up.

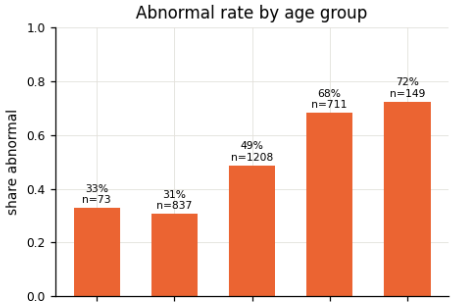
The **abnormal rate** of a group G of recordings (an age group, a sex, a session number, ...), with $y_i = 1$ for an abnormal and 0 for a normal recording, is

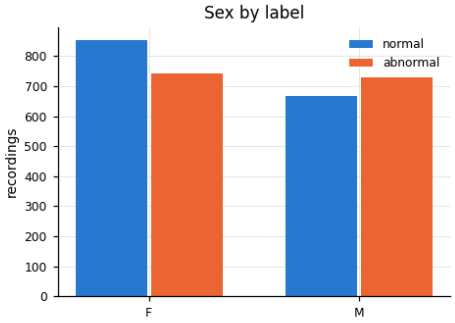
$$r(G) = \frac{1}{|G|} \sum_{i \in G} y_i. \tag{2.1}$$

label	count	mean	std	min	25%	50%	75%	age max
abnormal	1463.0	54.4	17.2	1.0	44.0	55.0	67.0	89.0
normal	1515.0	44.0	16.1	1.0	31.0	44.0	55.0	88.0

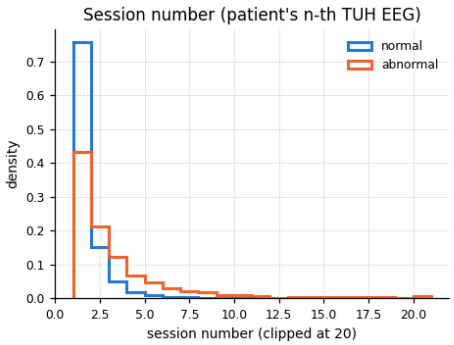
label share by sex	abnormal	normal
F	0.465	0.535
M	0.522	0.478

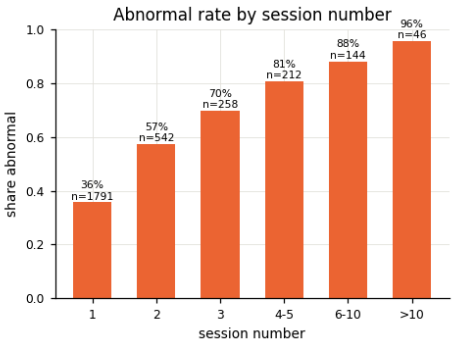


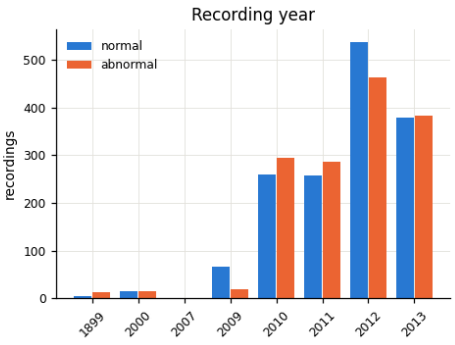





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subjects with >1 recording: 443 / 2329;  subjects with BOTH a normal and an abnormal re
cording: 54
```







3. Acquisition setup — real information or a shortcut?

These features describe *how* the EEG was recorded, not the patient. A feature like this is a **shortcut** when its abnormal rate differs strongly between values. A model can then learn the hospital workflow (which room, amplifier or protocol was used for sicker patients) instead of brain activity, and that will not transfer to another hospital. All of them are read from the EDF file, apart from `photic_performed`, which is read from the stimulator's signal.

feature	what it is	values in TUAB	what it can reveal
`sfreq`	**Sampling rate**: how many samples per second the amplifier stored for each EEG channel (Hz), from the EDF signal header.	250 Hz (93%), 256 Hz (6%), 512 Hz (1%)	Which amplifier / acquisition software was used. Preprocessing resamples everything to 200 Hz, but the original rate leaves faint traces (anti-alias filter edge, resampling artifacts).
`physical_max_uv`	**Amplifier input range**: the largest voltage the channel can record (± µV, EDF "physical maximum"). With 16-bit storage it also sets the resolution (range / 32,767 ≈ 0.15 µV per step).	±4,999.8 µV (93%), ±5,482.3 µV (7%)	The amplifier model. It maps one-to-one onto the sampling rate: 4,999.8 µV ⇔ 250 Hz.
`equipment_tag`	**Machine name** from the EDF recording field. TUH anonymises most of it to `X`; some files keep a tag (`Compact_32ch`, `Compact_64ch`) or masked characters (`#####`).	`X` (94%), `#####` (4%), other tags (2%)	The amplifier family, wherever the tag survived. Every non-`X` tag is on the 256/512 Hz system.
`n_signals`	**Channels stored in the file**: EEG plus auxiliary channels (EKG, EMG, eye, photic, trend channels, spare inputs). It depends on the headbox and montage configuration of the room.	36 (55%), 30 (34%), 27–35 (11%)	The recording room / configuration. Rare layouts (e.g. 27 channels, 81% abnormal) are almost a label on their own.
`has_emg`	An **EMG channel** (muscle electrode, e.g. chin or limb; used for sleep and movement work-ups).	yes 61%	Protocol. Shows no label difference (48% vs 50% abnormal).
`has_eog`	**Eye channels** LOC/ROC: electrodes beside the eyes that record eye movements, so eye artifacts can be told apart from frontal brain activity.	yes 93%	Missing exactly on the 256/512 Hz system, so it is the same amplifier shortcut.
`has_sphenoidal`	**Sphenoidal electrodes** SP1/SP2: thin wires placed near the skull base to record the deep (mesial) temporal lobe, used in epilepsy and epilepsy-surgery work-ups.	yes 3.5%	The **reason for referral** (suspected temporal-lobe epilepsy); 63% abnormal.
`has_numbered_extra`	**Spare amplifier inputs** stored as numbered channels (`26` … `32`) with nothing connected.	yes 59%	Headbox configuration only; no label difference (51% vs 47%).
`photic_performed`	**Photic stimulation**, detected from the stimulator channel. A strobe flashes at increasing rates during a routine EEG to provoke photosensitive responses; this flag says whether it was actually done.	done 72%	The **patient's state**: it is skipped in obtunded, ICU or uncooperative patients, so recordings without it are 80% abnormal vs 37% with it. The flashes also drive a visible rhythm in the occipital EEG, so the model *can* see whether it happened.

Four of these columns are one fact. `sfreq` ≠ 250 Hz, `physical_max_uv` = 5,482.3, a non-`X` `equipment_tag` and a missing EOG all mark the same second amplifier system. It covers 207 recordings (7%), 68% of them abnormal against 48% on the main system, and only 4% of them are in the eval split. Treat them as one shortcut, not four independent findings. For LaBraM this matters less, because it only sees the 19–23 EEG channels after resampling; but the **sampling rate and amplifier range survive preprocessing** as subtle spectral/amplitude fingerprints, and photic stimulation is visible in the EEG itself.

The table below lists, per value of each feature, the number of `recordings`, the `abnormal_rate` (share of those recordings labelled abnormal) and the `eval_share` (share of them in the official eval split). The colour scale is centred on the overall abnormal rate, so red = more abnormal than average and blue = less.

feature	value	recordings	abnormal_rate	eval_share
sfreq	250.000000	2786	48%	10%
sfreq	256.000000	189	71%	4%
sfreq	512.000000	18	33%	0%
physical_max_uv	4999.847000	2786	48%	10%
physical_max_uv	5482.288000	207	68%	4%
equipment_tag	#	28	82%	7%
equipment_tag	#####	116	60%	4%
equipment_tag	Compact_32ch	22	86%	0%
equipment_tag	Compact_64ch	14	64%	0%
equipment_tag	X	2813	48%	10%
n_signals	27	62	81%	8%
n_signals	28	29	55%	0%
n_signals	29	8	88%	12%
n_signals	30	1012	45%	7%
n_signals	31	139	44%	11%
n_signals	33	26	73%	4%
n_signals	34	72	60%	1%
n_signals	35	7	57%	0%
n_signals	36	1638	50%	11%
has_emg	0.000000	1179	48%	7%
has_emg	1.000000	1814	50%	11%
has_eog	0.000000	207	68%	4%
has_eog	1.000000	2786	48%	10%
has_sphenoidal	0.000000	2888	49%	9%
has_sphenoidal	1.000000	105	63%	2%
has_numbered_extra	0.000000	1219	47%	7%
has_numbered_extra	1.000000	1774	51%	11%
photic_performed	0.000000	837	80%	9%
photic_performed	1.000000	2156	37%	9%

4. Data quality — is the data good, noisy, or missing?

Computed per recording by `labram/data/tuh_meta_features.py::quality_features` over the 19 scalp electrodes $c = 1..C$. Notation: $x_c(t)$ is the raw signal (µV); $\tilde{x}_c(t)$ the same signal band-passed 0.5–45 Hz (4th-order Butterworth, zero-phase); the band-passed signal is cut into E non-overlapping 10-s epochs e; σ_{ec} and $\mathrm{ptp}_{ec} = \max - \min$ are the standard deviation and peak-to-peak range of $\tilde{x}_c$ in epoch e (µV).

- **Clipping** (saturation): the share of raw samples at ≥ 99% of the amplifier range $V_{\max}$ (`physical_max_uv`), over all T samples and channels:

$$\texttt{q_clip_frac} = \frac{1}{CT} \sum_{c,t} 1\big[|x_c(t)| \geq 0.99\, V_{\max}\big]. \tag{4.1}$$

- **Flat** and **high-amplitude** channel-epochs (a disconnected electrode; movement or an electrode pop):

$$\begin{aligned} \mathrm{flat}_{ec} &= 1[\sigma_{ec} < 0.5\,\mu\mathrm{V}] \\ \mathrm{high}_{ec} &= 1[\mathrm{ptp}_{ec} > 500\,\mu\mathrm{V}]. \end{aligned} \tag{4.2}$$

- **Epoch fractions**: an epoch counts when *any* electrode is affected:

$$\texttt{q_flat_epoch_frac} = \frac{1}{E}\sum_{e} \max_{c} \mathrm{flat}_{ec}$$
$$\texttt{q_high_amp_epoch_frac} = \frac{1}{E}\sum_{e} \max_{c} \mathrm{high}_{ec} \tag{4.3}$$

$$\texttt{q_bad_epoch_frac} = \frac{1}{E}\sum_{e} \max_{c} \left(\mathrm{flat}_{ec} \vee \mathrm{high}_{ec}\right). \tag{4.4}$$

- **Flat electrodes**: flat in more than half of the epochs, $\texttt{q_n_flat_channels} = \sum_c 1\left[\frac{1}{E}\sum_e \mathrm{flat}_{ec} > 0.5\right]$. **Noisy electrodes**: not flat, but with a whole-recording amplitude that is a robust outlier among the recording's electrodes. With $s_c = \ln(\mathrm{std}_t\, \tilde{x}_c + 10^{-6})$:

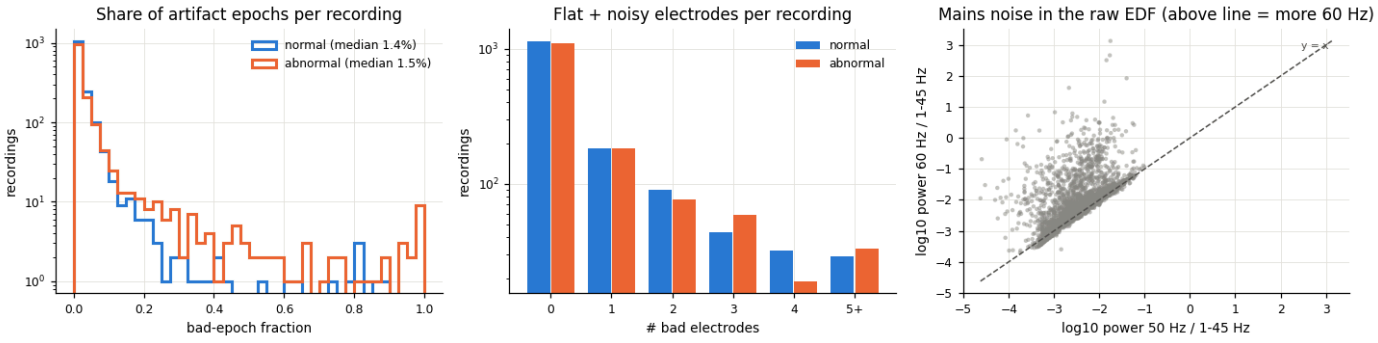
$$\texttt{q_n_noisy_channels} = \sum_c 1\left[\frac{|s_c - \mathrm{med}_{c'}\, s_{c'}|}{1.4826\left(\mathrm{MAD}_{c'}\, s_{c'} + 10^{-6}\right)} > 5\right]\left(1 - \mathrm{flatch}_c\right),$$

  where MAD is the median absolute deviation from the median, 1.4826 makes it a standard-deviation estimate for Gaussian data, and flatch_c marks a flat electrode.
- **Median amplitude**: $\texttt{q_median_amp_uv} = \mathrm{med}_{e,c}\ \sigma_{ec}$ (µV).
- **Mains noise**: with $\bar{P}(f)$ the Welch power spectrum of the *raw* signal (2-s Hann windows, 50% overlap) averaged over electrodes, the log-share of power within ±1 Hz of the mains frequency $f_0 \in \{50, 60\}$ Hz:

$$\texttt{q_line}f_0\texttt{_log_ratio} = \log_{10}\left(\frac{\sum_{|f-f_0|\le 1}\bar{P}(f)}{\sum_{1\le f\le 45}\bar{P}(f)} + 10^{-12}\right). \tag{4.6}$$

The **quality tier** used below (code cell after Figure "share of artifact epochs"), with b = `q_bad_epoch_frac`, n_{flat}, n_{noisy} the electrode counts and k = `q_clip_frac`:

$$\mathrm{tier} = \begin{cases} \text{poor} & n_{\mathrm{flat}} \ge 2 \ \vee\ b > 0.3 \ \vee\ k > 0.01 \\ \text{fair} & \text{else if } n_{\mathrm{flat}} + n_{\mathrm{noisy}} \ge 1 \ \vee\ b > 0.1 \\ \text{good} & \text{otherwise.} \end{cases} \tag{4.7}$$



tuh_split	label	quality_tier	good	fair	poor	All
eval	abnormal		92	30	4	126
	normal		106	41	3	150
train	abnormal		921	361	64	1346
	normal		997	359	15	1371
All			2116	791	86	2993

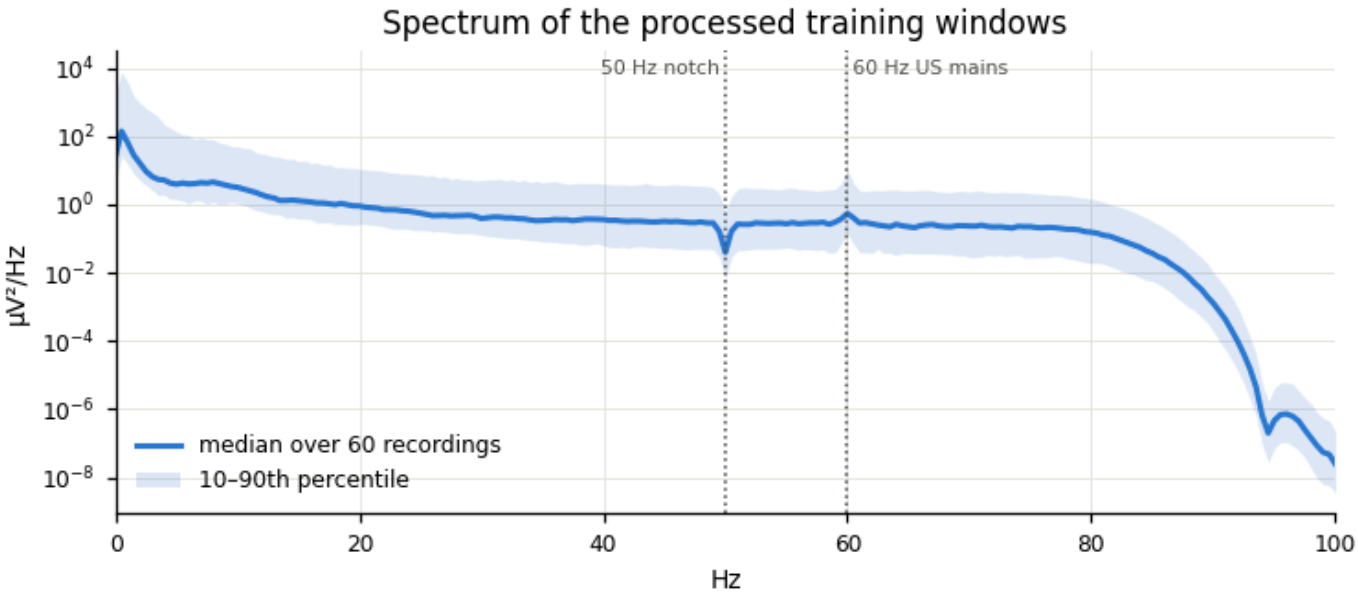
Worst recordings by artifact share:

	label	tuh_split	age	q_bad_epoch_frac	q_clip_frac	q_n_flat_channels	q_n_noisy_channels	q_median_amp_uv	duration_min
1392	abnormal	train	20.0	1.000	4.644e-03	0	3	36.371	29.400
1202	abnormal	train	34.0	1.000	0.000e+00	0	0	114.400	16.400
1186	abnormal	train	49.0	1.000	0.000e+00	0	0	68.063	23.917
1507	abnormal	train	29.0	1.000	1.925e-01	0	0	1731.404	18.083
65	abnormal	eval	43.0	0.996	0.000e+00	3	0	0.719	45.433
1231	abnormal	train	20.0	0.993	1.637e-04	0	0	38.081	24.750
1590	abnormal	train	42.0	0.990	0.000e+00	8	0	0.598	34.050
1222	abnormal	train	49.0	0.990	0.000e+00	6	0	0.635	50.500
1349	abnormal	train	51.0	0.977	2.865e-05	0	0	34.768	21.800
352	abnormal	train	71.0	0.975	7.012e-07	2	0	1.146	40.033

Does the preprocessing remove the mains noise?

`dataset_maker/make_TUAB.py` band-passes 0.1–75 Hz and applies a **50 Hz** notch. TUH was recorded in Philadelphia, where the mains frequency is **60 Hz** — so the notch misses it and the 75 Hz low-pass keeps it. Checked directly on a sample of the processed windows the model trains on:

```
60 Hz peak vs 55-58 Hz floor: median x2.2, >10x in 17% of sampled recordings
```



5. Why is a case abnormal?

TUAB does not say. The clinical reasons for an "abnormal" EEG read are, in rough order of frequency: **background slowing** (diffuse or focal), a **slow or absent posterior dominant rhythm**, **asymmetry**, **epileptiform discharges** (spikes/sharp waves), and rarer patterns (burst-suppression, low voltage). Most of these have a quantitative EEG proxy, so we can recover a *putative* reason per recording:

- each signature is a qEEG feature + the direction in which it becomes abnormal;
- the threshold is the 95th (or 5th) percentile of **normal training recordings in the same age group** — so ~5% of normals are flagged by construction, and age effects (the PDR slows with age) don't masquerade as abnormality.

These are *screening phenotypes*, not diagnoses — epileptiform discharges in particular are brief and rarely move a whole-recording statistic.

Are the signatures taken from medical records?

No. Every signature is **calculated**; none comes from a medical record:

	where it comes from
The signature names (diffuse slowing, focal slowing, slow / absent PDR, asymmetry, sharp transients, low voltage)	**Clinical EEG knowledge**: the usual reasons an EEG reader calls a recording abnormal. The categories, and the one qEEG feature chosen for each with the direction in which it becomes abnormal (`SIGNATURES` in `labram/eval/meta_analysis.py` , e.g. "diffuse slowing ↔ high slowing ratio"), are a rule written down for this analysis, not data.
Whether a recording has a signature	**Computed from its EEG signal.** The qEEG features of Eqs. (5.1)–(5.9) are computed by `labram/data/tuh_meta_features.py::qeeg_features` from each EDF. A recording is flagged when its feature lies beyond the 95th (or 5th) percentile of age-matched *normal training* recordings, Eqs. (5.13)–(5.14). The 95% level is a convention, so ≈5% of normals are flagged by design.
Medical records of TUAB	**None exist.** TUAB ships only the normal/abnormal folder label. TUH stopped distributing the EEG reports, which held the findings, diagnosis and medication.

So for TUAB a signature is a **hypothesis** about *why* a recording is abnormal, generated from the signal.

Medical-record information exists only for the **TUEG cohort** in Part II, through NEDC's clinical metadata sheet:

- the report's free-text **"abnormal due to"** reason (e.g. "generalized background slowing", "left temporal sharp waves");
- the report findings FIRDA / OIRDA / TIRDA / GPD / LPD;
- 15 diagnoses and the medication list.

Section 13 uses the "abnormal due to" reason as ground truth to **check** these signatures: how often a report that says "diffuse slowing" also gets the diffuse-slowing signature, and so on.

qEEG features (formulas)

`labram/data/tuh_meta_features.py::qeeg_features`. The scalp signals are re-referenced to the common average and band-passed 0.5–45 Hz (4th-order Butterworth, zero-phase):

$$\tilde{x}_c(t) = \mathrm{BP}_{0.5\text{–}45}\left[x_c(t) - \frac{1}{C}\sum_{c'} x_{c'}(t)\right]. \tag{5.1}$$

Each 10-s epoch e gets a Welch spectrum $P_{ec}(f)$ (2-s Hann windows, 50% overlap, 0.5 Hz bins); the recording's spectrum is the **median over epochs**, so short artifacts do not dominate, and a band power is the **sum of the bins** in $[f_{\mathrm{lo}}, f_{\mathrm{hi}})$:

$$
\begin{aligned}
P_c(f) &= \operatorname{med}_e P_{ec}(f) \\
B_c[f_{\mathrm{lo}}, f_{\mathrm{hi}}) &= \sum_{f_{\mathrm{lo}} \le f < f_{\mathrm{hi}}} P_c(f) \\
T_c &= B_c[1, 30).
\end{aligned}
\tag{5.2}
$$

With the bands $\delta = [1,4)$, $\theta = [4,8)$, $\alpha = [8,13)$, $\beta = [13,30)$ Hz and $\langle \cdot \rangle_c$ the mean over electrodes:

$$
\begin{aligned}
\texttt{qeeg_rel_}b &= \left\langle \frac{B_c[b]}{T_c} \right\rangle_c \\
\texttt{qeeg_log_total_power} &= \log_{10} \langle T_c \rangle_c
\end{aligned}
\tag{5.3}
$$

$$
\begin{aligned}
\texttt{qeeg_theta_alpha_ratio} &= \log_{10} \frac{\texttt{rel}_\theta}{\texttt{rel}_\alpha} \\
\texttt{qeeg_slowing_ratio} &= \log_{10} \frac{\texttt{rel}_\delta + \texttt{rel}_\theta}{\texttt{rel}_\alpha + \texttt{rel}_\beta}.
\end{aligned}
\tag{5.4}
$$

(The total power is a sum of 0.5-Hz bins of μV²/Hz, i.e. twice the power in μV²; only differences between recordings matter.)

Posterior dominant rhythm (PDR) and aperiodic slope. Average the spectra of the posterior electrodes $\mathcal{P} = \{O1, O2, P3, P4\}$, fit a straight line in log-log space over 2–30 Hz by ordinary least squares, and look for the largest bump above that line between 5 and 14 Hz:

$$
\begin{aligned}
\bar{P}(f) &= \frac{1}{|\mathcal{P}|} \sum_{c \in \mathcal{P}} P_c(f) \\
(a, \beta_{1/f}) &= \arg\min_{a,\beta} \sum_{2 \le f \le 30} \left(\log_{10} \bar{P}(f) - a - \beta \log_{10} f \right)^2
\end{aligned}
\tag{5.5}
$$

$$
\begin{aligned}
\rho(f) &= \log_{10} \bar{P}(f) - a - \beta_{1/f} \log_{10} f \\
\texttt{qeeg_pdr_freq} &= \arg\max_{5 \le f \le 14} \rho(f)
\end{aligned}
\tag{5.6}
$$

$$
\begin{aligned}
\texttt{qeeg_pdr_prominence} &= \rho(\texttt{qeeg_pdr_freq}) \\
\texttt{qeeg_aperiodic_slope} &= \beta_{1/f}.
\end{aligned}
\tag{5.7}
$$

The PDR frequency is in Hz on the 0.5-Hz grid; a prominence near 0 means no distinct rhythm.

Asymmetry, over the 8 homologous pairs (l, r) (Fp1–Fp2, F3–F4, C3–C4, P3–P4, O1–O2, F7–F8, T3–T4, T5–T6), with $\epsilon = 10^{-12}$:

$$
\begin{aligned}
\texttt{qeeg_max_asymmetry} &= \max_{(l,r)} \left| \log_{10} \frac{T_l}{T_r} \right| \\
\texttt{qeeg_max_delta_asymmetry} &= \max_{(l,r)} \left| \log_{10} \frac{B_l[\delta] + \epsilon}{B_r[\delta] + \epsilon} \right|.
\end{aligned}
\tag{5.8}
$$

Sharp transients: the median over electrodes of the excess (Fisher) kurtosis of the whole filtered trace, with μ_c, σ_c its mean and standard deviation:

$$\texttt{qeeg_median_kurtosis} = \operatorname{med}_c \left(\frac{\frac{1}{T}\sum_t (\tilde{x}_c(t) - \mu_c)^4}{\sigma_c^4} - 3 \right). \tag{5.9}$$

Auxiliary-channel features (formulas)

`ekg_features` and `auxiliary_features` in the same module.

- **Heart rate.** The EKG lead is band-passed 5–30 Hz and flipped if its 0.5th percentile is larger in magnitude than its 99.5th, so the R-peaks point up. R-peaks are the local maxima above $0.4\times$ its 99th percentile, at least 0.33 s apart, at times t_k. With the plausible intervals $\mathcal{K} = \{k : 0.33 < t_{k+1} - t_k < 2.0 \text{ s}\}$ (30–180 bpm):

$$\begin{aligned} \texttt{ekg_hr_bpm} &= \frac{60}{\operatorname{med}_{k\in\mathcal{K}}(t_{k+1} - t_k)} \\ \texttt{ekg_rr_std_ms} &= 1000 \cdot \operatorname{std}_{k\in\mathcal{K}}(t_{k+1} - t_k). \end{aligned} \tag{5.10}$$

  Both are NaN with fewer than 10 peaks or plausible intervals; `ekg_valid_frac` = $|\mathcal{K}|$ / all intervals.
- **Photic stimulation**: with $p(t)$ the stimulator channel and S the whole seconds of the recording, `photic_performed` = $1[\max p - \min p > 0]$ and

$$\texttt{photic_stim_sec} = \sum_{s=1}^{S} 1\big[\exists\, t \in \text{second } s : p(t) \neq 0\big]. \tag{5.11}$$

- **Device trends** (1-Hz channels SUPPR in %, BURSTS per min): the mean over the recording, and the share of seconds above 50% suppression; NaN when the trend is all zero (not computed):

$$\begin{aligned} \texttt{trend_suppr_mean} &= \frac{1}{S}\sum_s \mathrm{SUPPR}_s \\ \texttt{trend_suppr_gt50_frac} &= \frac{1}{S}\sum_s 1[\mathrm{SUPPR}_s > 50]. \end{aligned} \tag{5.12}$$

How the signature table is computed

`labram/eval/meta_analysis.py::signature_flags`, step by step:

1. **Each signature is one qEEG feature plus a direction** (`SIGNATURES`; the `feature` column):
   - *high* = abnormal: `diffuse_slowing` (slowing ratio, Eq. (5.4)), `focal_slowing` (max delta asymmetry, Eq. (5.8)), `asymmetry` (max total-power asymmetry, Eq. (5.8)),

`sharp_transients` (kurtosis, Eq. (5.9));
   - *low* = abnormal: `slow_pdr` (PDR frequency, Eq. (5.6)), `absent_pdr` (PDR prominence, Eq. (5.7)), `low_voltage` (log total power, Eq. (5.3)).
2. **Reference set** $\mathcal{R}$ = the **normal recordings of the train split** (1,371 recordings). Eval recordings never set a threshold.
3. **One threshold per age group** (<20, 20–39, 40–59, 60–79, 80+), from the reference recordings of that group, so that normal ageing (e.g. the PDR slowing) is not called abnormal:

$$\begin{aligned} \tau_{s,g} &= Q_{0.95}\big(\{x_{js} : j \in \mathcal{R},\ g(j) = g\}\big)\ \text{(high)} \\ \tau_{s,g} &= Q_{0.05}(\cdot)\ \text{(low)} \end{aligned} \tag{5.13}$$

   where x_{js} is signature s's feature for recording j, $g(j)$ its age group and Q_q the q-quantile (linear interpolation). A group with fewer than 20 reference recordings uses the pooled reference instead.
4. **Flag** recording i when it is beyond the threshold of its own age group (a missing value is never flagged):

$$\begin{aligned} f_{is} &= 1\big[x_{is} > \tau_{s,g(i)}\big]\ \text{(high)} \\ f_{is} &= 1\big[x_{is} < \tau_{s,g(i)}\big]\ \text{(low)}. \end{aligned} \tag{5.14}$$

5. **The table's columns**, over **all** recordings of both splits:

$$\begin{aligned} \text{normal}_s &= \frac{1}{|N|}\sum_{i \in N} f_{is} \\ \text{abnormal}_s &= \frac{1}{|A|}\sum_{i \in A} f_{is} \\ \text{lift}_s &= \frac{\text{abnormal}_s}{\text{normal}_s} \end{aligned} \tag{5.15}$$

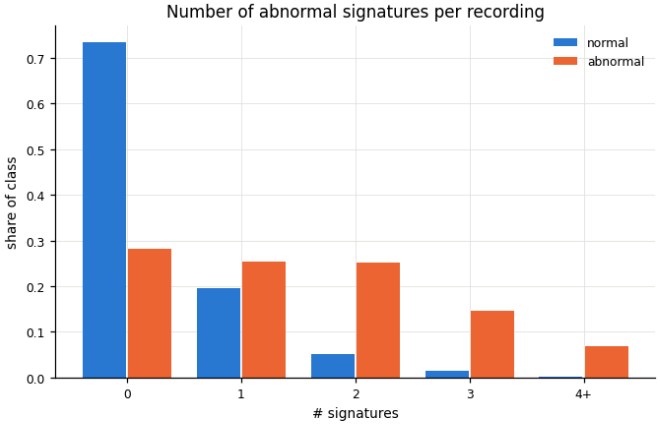
   with N the 1,521 normal and A the 1,472 abnormal recordings.

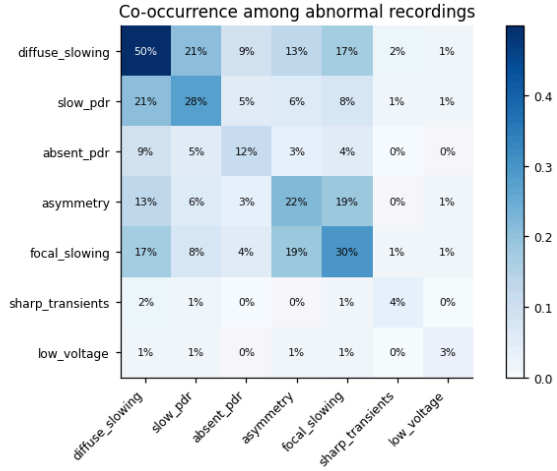
Reading it. `normal` ≈ 5% by construction: it is exactly 5.1% on the train normals that set the thresholds. It drifts on the 150 eval normals, which are included in the column but did not set the thresholds. `slow_pdr` sits lower (3.8%) because the PDR frequency comes in 0.5 Hz steps, and the strict "<" in Eq. (5.14) does not flag recordings that tie at the threshold. `lift` says how many times more often abnormal recordings carry the signature: ×9.8 for diffuse slowing is a strong marker. A lift **below 1** (sharp transients, low voltage) means the signature fires *more* often in normals, so that proxy does not work (see the findings below).

The *dominant* signature of a flagged recording (the `putative_reason` table) is the one with the largest exceedance in reference standard deviations, $z_{is} = \pm\,(x_{is} - \text{med}_{j \in \mathcal{R}}\, x_{js}) / \text{std}_{j \in \mathcal{R}}\, x_{js}$ (sign $+$ for *high*, $-$ for *low* signatures), taken over the signatures with $f_{is} = 1$.

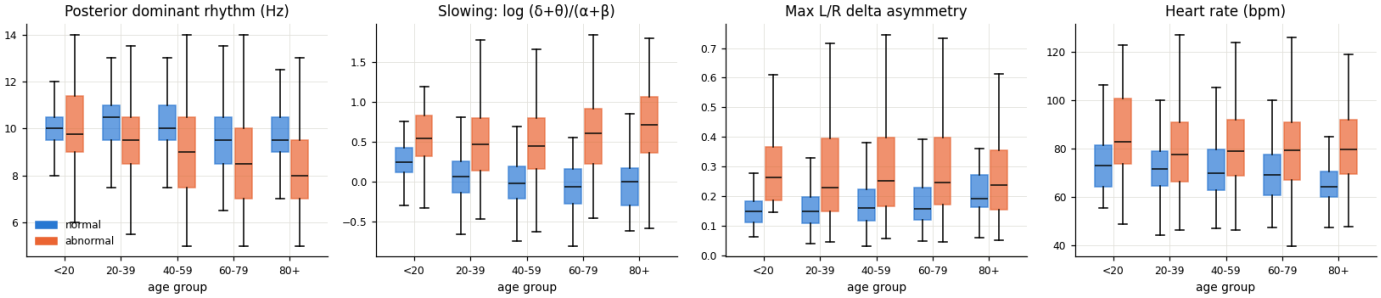
	normal	abnormal	lift	feature
diffuse_slowing	5.1%	49.9%	x9.8	qeeg_slowing_ratio
slow_pdr	3.8%	27.9%	x7.3	qeeg_pdr_freq
focal_slowing	5.3%	29.7%	x5.6	qeeg_max_delta_asymmetry
asymmetry	5.5%	21.9%	x4.0	qeeg_max_asymmetry
absent_pdr	4.8%	11.5%	x2.4	qeeg_pdr_prominence
sharp_transients	5.6%	3.7%	x0.7	qeeg_median_kurtosis
low_voltage	5.4%	2.9%	x0.5	qeeg_log_total_power

abnormal recordings with NO signature: 28% (normals with none: 73%)





label putative_reason	abnormal	normal	abnormal %
diffuse_slowing	519	66	35.3
none detected	414	1117	28.1
focal_slowing	264	64	17.9
slow_pdr	117	43	7.9
asymmetry	76	63	5.2
sharp_transients	36	77	2.4
low_voltage	28	60	1.9
absent_pdr	18	31	1.2



Clinical-context hints hidden in the setup

TUAB has no referral reason, but some recording details are indirect evidence of *why* and *in what state* the patient was recorded. Each row of the table below is one condition that can be read off a recording:

condition	how it is read	what it hints at
sphenoidal electrodes	SP1/SP2 channels exist in the EDF (`has_sphenoidal`)	Fine wire electrodes near the skull base reach the deep temporal lobe. They are placed when **temporal-lobe epilepsy** is suspected, i.e. in an epilepsy work-up.
session number > 5	the recording comes from the patient's 6th or later TUH EEG session (`session_number` , Section 2)	A patient recorded many times: **chronic neurological follow-up** (epilepsy, recurrent ICU stays).
suppression > 50% for > 5% of record	among recordings whose amplifier wrote the `SUPPR` trend channel, more than 5% of the seconds have a value above 50 (`trend_suppr_gt50_frac` , Eq. (5.12))	*Expected*: sustained burst-suppression, i.e. an **ICU / encephalopathy / anaesthesia** context. *Observed*: the opposite (see below).

condition	how it is read	what it hints at
photic stimulation done / not done	the photic stimulator channel carried flashes (`photic_performed` , Eq. (5.11))	Photic stimulation is a routine step of an outpatient EEG. Skipping it suggests a patient who **could not cooperate**: obtunded, sedated, ICU, or a short bedside recording.

Columns (all over the 2,993 TUAB recordings):

- `recordings` : recordings that meet the condition.
- `abnormal_rate` : their abnormal rate $r(G)$, Eq. (2.1). For comparison, the base rate of all recordings is in the last row.
- `abnormal_rate_rest` : the abnormal rate of the recordings where the condition is known to be **false**. Recordings where it cannot be evaluated are left out: the suppression row compares only recordings *with* a `SUPPR` trend.
- `lift` = `abnormal_rate` / `abnormal_rate_rest` .
- `mean_age` : the mean age (years) of the recordings meeting the condition, since several conditions go with older patients.

	recordings	abnormal_rate	abnormal_rate_rest	mean_age	lift
sphenoidal electrodes	105	63%	49%	49.1	x1.29
session number > 5	190	90%	46%	50.8	x1.94
suppression > 50% for > 5% of record	1,319	25%	54%	47.1	x0.46
no SUPPR trend written at all	1,187	74%	33%	51.2	x2.28
photic stimulation done	2,156	37%	80%	46.8	x0.47
no photic stimulation	837	80%	37%	55.1	x2.15
all recordings (base rate)	2,993	49%	–	49.1	–

Reading the table, row by row:

- **Sphenoidal electrodes**: 63% abnormal vs 49% without them (×1.3), at the same mean age. An epilepsy work-up is more often read abnormal, but most of these patients are still normal between seizures.
- **Session number > 5**: **90%** abnormal vs 46%. Patients who keep coming back are chronically ill; the session number is a strong proxy for illness, not for brain activity.
- **Suppression > 50%**: only **25%** abnormal vs 54% for the other recordings with a trend. A sustained clinical burst-suppression pattern would be almost always abnormal, so this device trend cannot mean what its name says. Its median is 68% in normal vs 30% in abnormal recordings. Its semantics are unknown (perhaps the device's own "quiet EEG" measure), so it must not be read clinically.
- **No SUPPR trend written**: 74% abnormal. Whether the trend exists depends on the amplifier and software (Section 3), so this is a recording-room shortcut, not physiology.
- **Photic stimulation**: skipped in 837 recordings, **80%** of them abnormal and 8 years older on average, against 37% when it was done. This is the clearest workflow signal: a patient who could not do photic stimulation was usually sicker.

None of these are measurements of the brain. They are useful to **stratify the evaluation** (does a model still separate normal from abnormal *within* photic yes/no?), but should not be model inputs (Section 7).

6. Can the meta features be used for training or

prediction?

The idea. LaBraM predicts normal vs abnormal from raw EEG windows. This section asks how far one gets **without the raw EEG**: from the per-recording meta features of this notebook alone (age, sex, setup, quality, heart rate, and the qEEG spectral summaries). The meta-feature model is a separate, cheap classifier that is **never fed into LaBraM**. It serves three purposes:

1. **A baseline LaBraM has to beat.** If 13 spectral numbers plus age already reach a given AUROC, LaBraM's raw-EEG model is only worth its cost above that.
2. **A shortcut detector.** If *acquisition* or *history* features alone predict the label, the label is entangled with the workflow (Sections 3 and 5), and a model can cheat.
3. **A guide to fusion.** Features that are legitimately known at prediction time and carry information LaBraM may miss (age, sex) are candidates to add to the model (Section 7).

Concretely: one row per **recording**, target $y = 1$ for abnormal. The model is trained on the TUAB **train** recordings and evaluated on the official, subject-disjoint TUAB **eval** recordings, once per feature group, so each group's contribution can be read separately.

Two questions:

1. **How much label information is in each feature on its own?** (univariate ROC-AUC on the training split, 0.5 = none)
2. **How well does metadata *alone* predict normal/abnormal?** Gradient-boosted trees trained on TUAB train, scored on the official, subject-disjoint TUAB eval set, one feature group at a time.

> **Comparing with LaBraM.** The published LaBraM-Base TUAB numbers (AUROC 0.902, balanced accuracy 0.814) are **per 10-s window**. The baselines here are **per recording**, which is an easier unit: a whole-recording statistic averages out noise that a single window cannot. For a like-for-like comparison, use the recording-level (case-aggregated) LaBraM metrics from `notebooks/finetune_evaluation.ipynb`.

Metrics. For a score u_i (one feature, or the model's predicted probability p_i) and labels y_i, the ROC-AUC is the probability that a random abnormal recording scores higher than a random normal one (ties count half):

$$\mathrm{AUC}(u) = \frac{1}{|A|\,|N|} \sum_{i \in A} \sum_{j \in N} \Big(1[u_i > u_j] + \tfrac{1}{2}\, 1[u_i = u_j] \Big). \tag{6.1}$$

The **univariate** AUC of a feature is oriented so that 0.5 = no information and 1 = perfect, whichever direction separates (`labram/eval/meta_analysis.py::univariate_auc` ; missing values dropped per feature, `coverage` = share kept):

$$\begin{aligned} \mathrm{AUC}_{\mathrm{uni}}(x) &= \max\big(\mathrm{AUC}(x),\ 1-\mathrm{AUC}(x)\big) \\ \mathrm{direction} &= \begin{cases} \text{higher} = \text{abnormal} & \mathrm{AUC}(x) \geq 0.5 \\ \text{lower} = \text{abnormal} & \text{else.} \end{cases} \end{aligned} \tag{6.2}$$

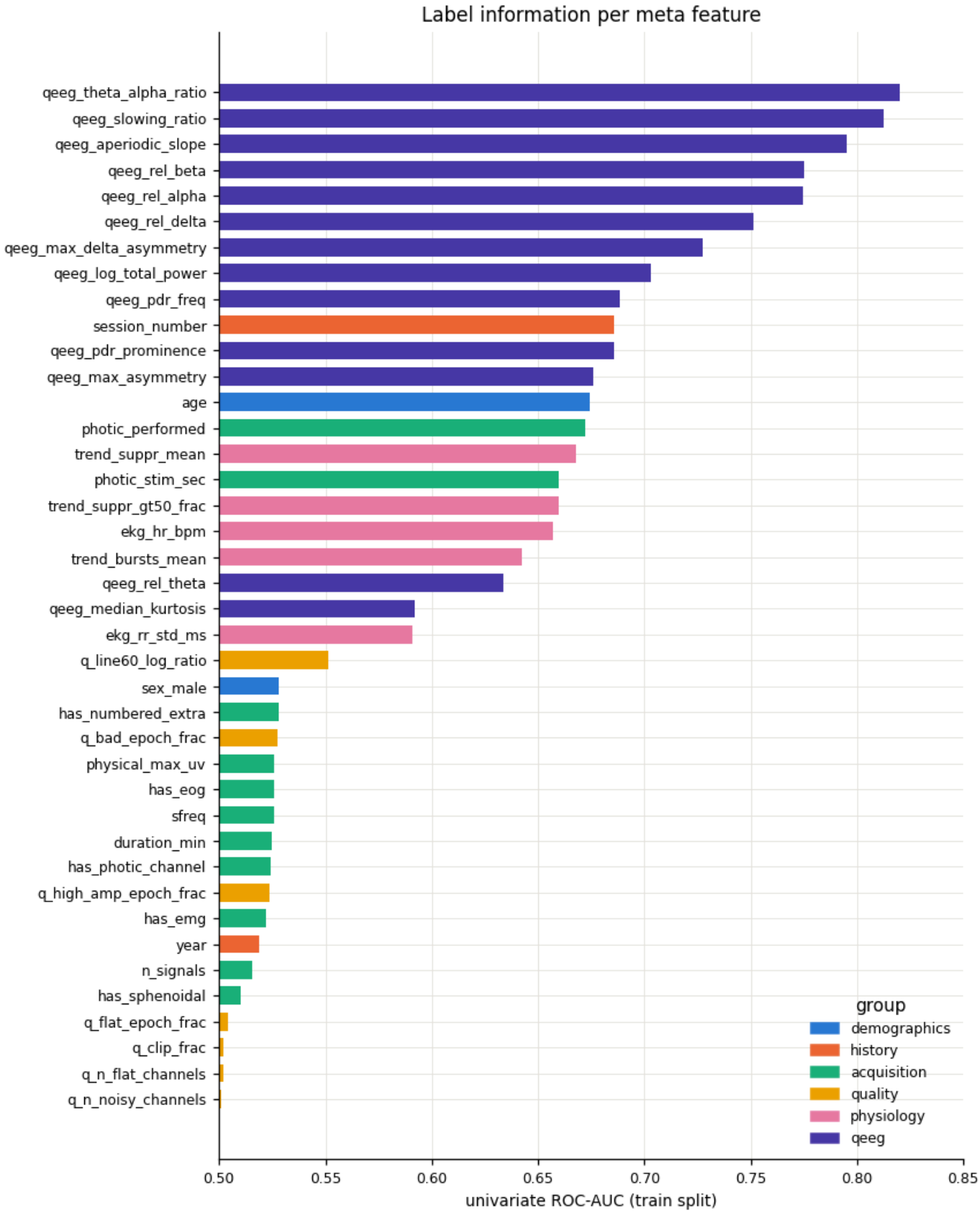
The metadata-only model (`metadata_baseline` : histogram gradient-boosted trees, 300 iterations, learning rate 0.05, 15 leaves, L2 = 1; NaNs handled natively) is scored by the AUROC of p_i, Eq. (6.1), and by the balanced accuracy at threshold 0.5:

$$\mathrm{BalAcc} = \frac{1}{2}\left(\frac{\sum_{i\in A}\mathbb{1}[p_i \geq 0.5]}{|A|} + \frac{\sum_{i\in N}\mathbb{1}[p_i < 0.5]}{|N|}\right). \tag{6.3}$$

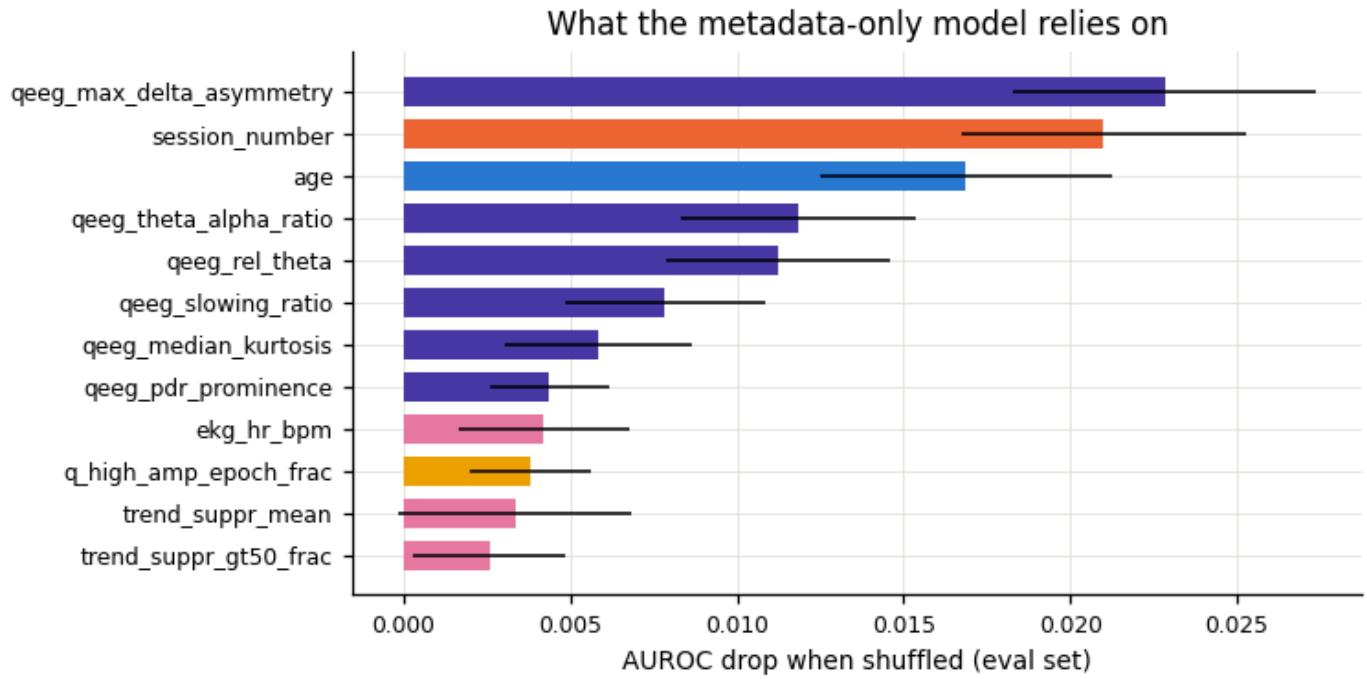
The **permutation importance** of feature k is the drop in eval AUROC when its column is shuffled, averaged over $R = 10$ shuffles π_r:

$$\Delta_k = \mathrm{AUC}(p) - \frac{1}{R}\sum_{r=1}^{R}\mathrm{AUC}\left(p^{(k\leftarrow\pi_r)}\right). \tag{6.4}$$

	feature	group	auc	direction	coverage	median_normal	median_abnormal
0	qeeg_theta_alpha_ratio	qeeg	0.820	higher=abnormal	1.000	-0.230	0.201
1	qeeg_slowing_ratio	qeeg	0.813	higher=abnormal	1.000	0.020	0.518
2	qeeg_aperiodic_slope	qeeg	0.795	lower=abnormal	1.000	-1.273	-1.892
3	qeeg_rel_beta	qeeg	0.775	lower=abnormal	1.000	0.184	0.092
4	qeeg_rel_alpha	qeeg	0.774	lower=abnormal	1.000	0.255	0.119
5	qeeg_rel_delta	qeeg	0.751	higher=abnormal	1.000	0.353	0.551
6	qeeg_max_delta_asymmetry	qeeg	0.728	higher=abnormal	1.000	0.155	0.248
7	qeeg_log_total_power	qeeg	0.703	higher=abnormal	1.000	1.761	2.012
8	qeeg_pdr_freq	qeeg	0.688	lower=abnormal	1.000	10.000	9.000
9	session_number	history	0.685	higher=abnormal	1.000	1.000	2.000
10	qeeg_pdr_prominence	qeeg	0.685	lower=abnormal	1.000	0.719	0.417
11	qeeg_max_asymmetry	qeeg	0.676	higher=abnormal	1.000	0.153	0.217
12	age	demographics	0.674	higher=abnormal	0.995	44.000	55.000
13	photic_performed	acquisition	0.672	lower=abnormal	1.000	1.000	1.000
14	trend_suppr_mean	physiology	0.668	lower=abnormal	0.598	69.328	30.386


Label information per meta feature
qeeg_theta_alpha_ratio
qeeg_slowing_ratio
qeeg_aperiodic_slope
qeeg_rel_beta
qeeg_rel_alpha
qeeg_rel_delta
qeeg_max_delta_asymmetry
qeeg_log_total_power
qeeg_pdr_freq
session_number
qeeg_pdr_prominence
qeeg_max_asymmetry
age
photic_performed
trend_suppr_mean
photic_stim_sec
trend_suppr_gt50_frac
ekg_hr_bpm
trend_bursts_mean
qeeg_rel_theta
qeeg_median_kurtosis
ekg_rr_std_ms
q_line60_log_ratio
sex_male
has_numbered_extra
q_bad_epoch_frac
physical_max_uv
has_eog
sfreq
duration_min
has_photic_channel
q_high_amp_epoch_frac
has_emg
year
n_signals
has_sphenoidal
q_flat_epoch_frac
q_clip_frac
q_n_flat_channels
q_n_noisy_channels
group
demographics
history
acquisition
quality
physiology
qeeg
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
univariate ROC-AUC (train split)

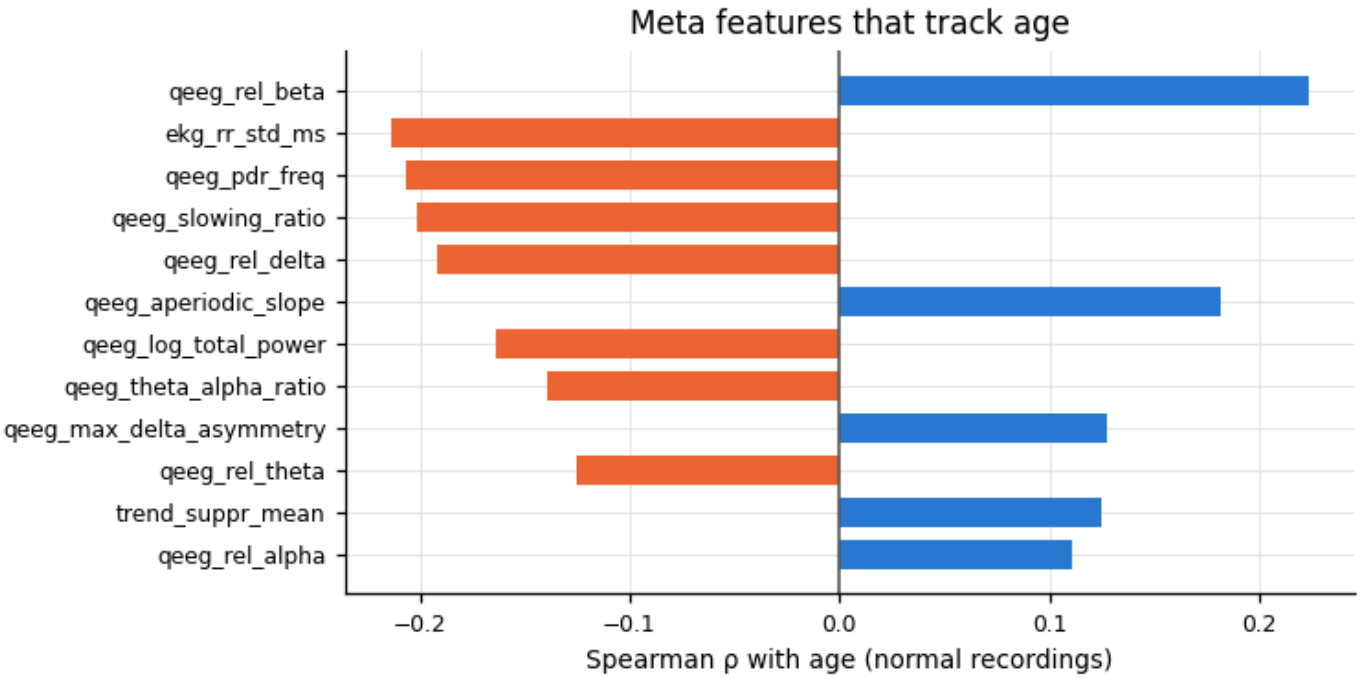
features	auroc	balanced_acc	n_features
demographics (age, sex)	0.689	0.639	2
+ history (session no., year)	0.727	0.644	4
acquisition only	0.686	0.658	11
quality only	0.574	0.575	7
physiology only (EKG, trends)	0.783	0.721	5
qEEG only	0.890	0.813	13
demographics + qEEG	0.901	0.832	15
all meta features	0.922	0.863	40
all except acquisition/history	0.907	0.831	27



Relevance for the age-regression task

For the brain-age task (`TUAB_AGE`), the interesting question is the reverse: which meta features *track age* — these are what a brain-age model will implicitly learn, and the non-EEG ones (heart rate, session number) are covariates to control for. Spearman ρ within **normal** recordings only, so pathology doesn't drive it, where Spearman's ρ is the Pearson correlation of the ranks $R(\cdot)$ (average ranks for ties), over the normal recordings with both values:

$$\rho(x, a) = \frac{\operatorname{cov}\big(R(x), R(a)\big)}{\sigma_{R(x)}\,\sigma_{R(a)}}. \tag{6.5}$$



7. Findings and recommendations

Numbers are from the full TUAB v3.0.0 sweep (2,993 recordings, 2,329 subjects, 0 extraction failures).

What meta information exists

	coverage	note
Age	99.5%	15 unusable: 12 × `Age:999` (90+, redacted), 3 × `Age:0`
Sex	100%	
Session number, recording year	100%	year **1899** (16 recordings) and **2000** (29) are placeholder dates; treat as missing
Heart rate / HRV (EKG lead)	100%	an EKG lead exists on every recording; R-peaks detected in all
Photic stimulation	95% have the channel; performed in 72%	
Device burst-suppression trends	60%	computed only by some acquisition systems
EMG / sphenoidal electrodes	61% / 3.5%	
Reason for abnormality, diagnosis, medication, vigilance	**0%**	not distributed by TUH

Demographics

- Abnormal patients are **~10 years older** (mean 54.4 vs 44.0). The abnormal rate climbs from 31% (age 20–39) to 72% (80+).
- Sex is nearly balanced: 52% of male recordings are abnormal, vs 47% of female ones.
- 443 subjects have more than one recording, and **54 have both a normal and an abnormal one**.
- **Session number is a strong proxy for illness.** 36% of first EEGs are abnormal, rising to 81% at sessions 4–5 and 96% after more than 10 sessions.

Data quality

- **The data is mostly clean.** Clipping is essentially absent, and the median recording has 1.4% artifact epochs. Quality tiers: **2,116 good / 791 fair / 86 poor**.
- **Poor quality concentrates in abnormal recordings.** Train has 64 poor abnormal recordings vs 15 poor normal ones; very ill and ICU patients produce more artifact.
- A handful of recordings are essentially unusable: 100% artifact epochs, or 6–8 dead electrodes (a few train recordings and one *eval* recording; cell 14 lists them when run locally). All of them are labelled abnormal, so every window they contribute teaches "noise = abnormal".
- **Mains noise is not removed.** `make_TUAB.py` notches 50 Hz, but TUH is US data with 60 Hz mains. In the processed training windows the 60 Hz peak is a median ×2.2 above the neighbouring floor, and over ×10 in 17% of recordings.
- 3 abnormal train EDFs never made it into the processed windows.

Why is a case abnormal? (putative, from qEEG)

signature	abnormal	normal
diffuse slowing	**50%**	5%
focal (delta) asymmetry	**30%**	5%
slow posterior rhythm	**28%**	4%
hemispheric asymmetry	**22%**	6%
absent posterior rhythm	12%	5%
sharp transients (kurtosis), low voltage	3–4%	5–6% (no signal)

- **72% of abnormal recordings show at least one signature**, vs 27% of normals. Many show several: slowing and slow PDR co-occur, and so do focal slowing and asymmetry.
- **28% of abnormal recordings show none.** These are most likely epileptiform-only abnormalities (brief spikes that whole-recording spectra cannot see), subtle findings, or label noise. They should be the hardest cases for any model.
- The kurtosis "spike" proxy does **not** work: kurtosis is *higher* in normals, because it is dominated by eye blinks and alpha bursts. Recovering epileptiform reasons needs event annotations (TUEV) or a TUEV-trained detector.
- **Context hints:** recordings *without* photic stimulation are **80% abnormal** (837 recordings), vs 37% with it. Sphenoidal electrodes mean 63% abnormal. More than 5 sessions means 90% abnormal.
- The device `SUPPR` trend is *higher* in normals (median 68% vs 30%). It is therefore **not** a clinical burst-suppression percentage; its semantics are unknown, so do not interpret it clinically.

Can the meta features be used? Metadata-only model, recording level, official eval set

features	AUROC	bal. acc
demographics (age, sex)	0.689	0.639
+ history (session number, year)	0.727	0.644
acquisition only	0.686	0.658
quality only	0.574	0.575
physiology only (EKG, trends)	0.783	0.721
qEEG only	**0.890**	**0.813**

features	AUROC	bal. acc
demographics + qEEG	0.901	0.832
patient-level (all except acquisition/history)	0.907	0.831
all meta features	0.922	0.863

13 hand-crafted spectral summaries plus age already recover most of the TUAB label. In practice the label mostly means "background slowing or asymmetry beyond what is normal for this age". That makes a **strong, cheap baseline** that LaBraM has to beat at the *recording* level.

Recommendations

1. **Use age (and sex) as inputs.** They are legitimately known at prediction time, and normal EEG ranges are age-dependent. The simplest route is late fusion: concatenate them to the pooled embedding before the head, or stack them on the case-level LaBraM score.
2. **Run a stacking experiment:** a gradient-boosted model on LaBraM's recording-level probability + qEEG + age. If it beats LaBraM alone, the encoder is missing information. If not, LaBraM has already absorbed it.
3. **Do not feed acquisition/history features to the model** (photic performed, session number, device tags, trend availability, sampling rate). They encode Temple's referral workflow and will not transfer to another hospital. **Do use them to stratify evaluation**: check LaBraM within photic yes/no and first-session vs repeat subgroups. A large drop within strata means the model learned the workflow.
4. **Use the putative reasons for error analysis and auxiliary targets.** Measure recall per phenotype, especially on the 28% of abnormals with no signature. Regressing slowing ratio / PDR as an auxiliary head is a cheap multi-task regularizer.
5. **Clean the data.** Drop or down-weight the 86 poor-tier recordings, or mask their bad epochs. Add a 60 Hz notch: this needs a reprocessing pass, and should be checked against the pre-trained checkpoint, which saw the same 60 Hz residue. Treat years 1899/2000 as missing.
6. **To get real abnormality reasons,** request the annotated TUH subcorpora (TUEV events, TUSZ seizures, TUSL slowing, TUEP epilepsy, TUAR artifacts) and join them to TUAB by subject id. They are not on disk or in S3 today, and the id overlap still has to be verified once they are downloaded.

Part II — the TUEG cohort (`all_both_v3`)

Part I describes TUAB: 2,993 recordings, a binary label and nothing else. The brain-age models are now trained on a much larger TUH cohort, **`all_both_v3`**: 29,309 recordings of 13,801 patients aged 18–89. It holds every TUEG recording with the standard channels (T1/T2 may be missing), plus the TUAB recordings. The patient-level split comes from `dataset_maker/make_cohort_dataset.py`: TUAB patients keep their TUAB age split, the TUAB eval set is the test set, and other TUEG patients follow the shared TUEG patient split.

Part II repeats the Part I questions on this cohort and adds one that TUAB cannot answer. NEDC's clinical sheet gives diagnoses, medications and the report's own *reason* for an abnormal read.

8. The cohort and its meta sources

TUEG is on local disk only as processed per-recording npy files; its 1.75 TB of EDFs are in S3. The meta table is therefore built from the read-only sources in Table 8.1, not from an EDF sweep:

```
python -m dataset_maker.make_cohort_meta \
    --cohort /data/datasets/EEG-public/TUH_EEG/cohorts/all_both_v3 --n_jobs 8
# -> <cohort>/meta_features.parquet  (~80 min on 8 cores, resumable)
```

Table 8.1. Sources of TUEG meta information

Source	Columns	Notes
cohort (`cohort.parquet` , `manifest.json` , `age_metadata.json`)	`split` , `age` , `sex` , `label` , `n_windows` , `duration_sec`	`label` = TUAB folder label if the recording is in TUAB, else the NEDC report (`abnormal due to` non-empty); none when neither exists
recording index (`dataset_maker/tuh_index.py`)	`session_number` , `year` , `montage` , `sfreq` , `missing_channels` , `corpora`	from the EDF headers, scanned while downloading
NEDC clinical sheet (`dataset_maker/tuh_clinical.py`)	`dx_*` (15 diagnoses), `med_*` (6 drug classes), `f_*` (FIRDA/OIRDA/TIRDA/GPD/LPD), `abnormal_due_to`	per session; **not in TUAB**; a session missing from the sheet has unknown (NaN) flags, never False
TUEG header dump (`DOCS/headers.tar.gz` , `tuh_meta_features.parse_header_dump`)	`equipment_family` , `physical_max_uv` , `has_ekg/emg/eog/photic_channel/sphenoidal/trend_channels`	the full EDF header of all 70,831 TUEG files, so no EDF has to be read
processed npy (`tuh_meta_features.npy_signal_features`)	`q_*` quality, `qeeg_*`	the same functions as Part I, on the 0.1–75 Hz, 200 Hz, 50/60 Hz-interpolated signal, first ≤ 60 min, recorded scalp channels only

Part I's EKG, photic and burst-suppression *signal* features are missing here, because those channels are not kept in the npy. Clipping and the mains ratio cannot be measured after band-pass, resampling and line removal, so they are dropped too.

```
29,309 recordings, 13,801 subjects, 110 columns; signal-feature failures: 0, not in the header dump: 0, not in the clinical sheet: 2,490
```

Table 8.2. Recordings per split and label

Recordings (subjects in the last column) of `all_both_v3` by split and label. *Unlabelled* recordings are TUEG sessions that are neither in TUAB nor in the clinical sheet. They train the age model but take no part in any normal/abnormal statistic below.

label3 split	normal	abnormal	unlabelled	all	subjects	from TUAB
train	10067	13714	1832	25613	12017	1868
val	1388	1676	358	3422	1533	482
test	149	125	0	274	251	274
all	11604	15515	2190	29309	13801	2624

Table 8.3. Coverage of the meta features, TUEG cohort vs TUAB

For every feature of the cohort groups (`labram/eval/meta_analysis.py::COHORT_FEATURE_GROUPS`), the coverage is the share of recordings with a value,

$$\mathrm{cov}(x) = \frac{1}{N}\sum_{i=1}^{N} 1\left[x_i \text{ not missing}\right], \tag{8.1}$$

over the N recordings of the cohort (`TUEG %`) and of Part I's TUAB table (`TUAB %`; "–" means the column does not exist there). `type`, `n_categories`, `categories` (the 5 most frequent values with their share), `active_range` (1st–99th percentile) and `meaning` (for the history, acquisition and qEEG groups) are defined as in Section 1.

group	feature	type	TUEG %	TUAB %	n_categories	categories	active_range	meaning
demographics	age	quantitative	100.0	99.5	–		19 – 87	
	sex_male	categorical	99.8	100.0	2	0 (52%), 1 (48%)		
history	session_number	quantitative	100.0	100.0	–		1 – 13	Which EEG session of the patient this is: n-th TUH EEG session, in date order over the whole TUH archive (s003 -> 3). NOT the patient's total number of sessions
	year	quantitative	100.0	100.0	–		2000 – 2016	Recording year (the date is anonymised to 01-JAN-YYYY); 1899 and 2000 are placeholders
acquisition	duration_min	quantitative	100.0	100.0	–		5 – 486	Length of the recording [min]
	sfreq	categorical	100.0	100.0	5	250 (60%), 256 (30%), 1000 (7%), 512 (2%), 400 (1%)		Native sampling rate of the EEG channels [Hz]; set by the amplifier / acquisition software
	physical_max_uv	categorical	100.0	100.0	4	5000 (37%), 5482 (32%), 5000 (30%), 29483 (1%)		Amplifier input range from the EDF header [±µV]; identifies the amplifier family
	n_signals_header	quantitative	100.0	–	–		27 – 41	Channels stored in the EDF (EEG + auxiliary); a montage / amplifier fingerprint
	has_ekg	categorical	100.0	100.0	1	1 (100%)		An EKG (heart) lead was recorded
	has_emg	categorical	100.0	100.0	2	0 (65%), 1 (35%)		An EMG (muscle) channel was recorded
	has_eog	categorical	100.0	100.0	2	0 (55%), 1 (45%)		Both eye-movement channels (LOC, ROC) were recorded
	has_photic_channel	categorical	100.0	100.0	2	0 (59%), 1 (41%)		The photic-stimulator trigger channel is present
	has_sphenoidal	categorical	100.0	100.0	2	0 (72%), 1 (28%)		Sphenoidal electrodes (SP1/SP2), placed for temporal-lobe epilepsy work-ups
	has_numbered_extra	categorical	100.0	100.0	2	1 (62%), 0 (38%)		Spare numbered amplifier inputs are stored (e.g. '26'...'32')
	has_trend_channels	categorical	100.0	–	2	1 (69%), 0 (31%)		The device wrote burst-suppression trend channels (SUPPR/BURSTS/IBI)
	montage_le	categorical	100.0	–	2	0 (70%), 1 (30%)		Linked-ears reference montage (02_tcp_le) instead of average reference
	montage_reduced	categorical	100.0	–	2	0 (100%), 1 (0%)		Reduced channel-set montage (*_a)
	n_missing_channels	quantitative	100.0	–	–		0 – 2	Standard 10-20 channels absent from the EDF (in practice T1/T2)
	equip_compact_32ch	categorical	100.0	–	2	0 (93%), 1 (7%)		Amplifier family from the EDF machine tag: Compact 32-channel
	equip_compact_64ch	categorical	100.0	–	2	0 (95%), 1 (5%)		Amplifier family from the EDF machine tag: Compact 64-channel
	equip_intution_ultrasom	categorical	100.0	–	2	0 (80%), 1 (20%)		Amplifier family from the EDF machine tag: Intuition/Ultrasom
	equip_alliance	categorical	100.0	–	2	0 (90%), 1 (10%)		Amplifier family from the EDF machine tag: Alliance
	equip_bravo	categorical	100.0	–	2	0 (99%), 1 (1%)		Amplifier family from the EDF machine tag: Bravo (400 Hz, 2002-2003)
	equip_b545	categorical	100.0	–	1	0 (100%)		Amplifier family from the EDF machine tag: B545
quality	q_flat_epoch_frac	quantitative	100.0	100.0	–		0 – 0.85	
	q_high_amp_epoch_frac	quantitative	100.0	100.0	–		0 – 1	
	q_bad_epoch_frac	quantitative	100.0	100.0	–		0 – 1	
	q_n_flat_channels	quantitative	100.0	100.0	–		0 – 1	

group	feature	type	TUEG %	TUAB %	n_categories	categories	active_range	meaning
	q_n_noisy_channels	quantitative	100.0	100.0	–		0 – 5	
	q_median_amp_uv	quantitative	100.0	100.0	–		1.37 – 50.5	
	dx_electrographic_seizure	categorical	91.5	–	2	0 (99%), 1 (1%)		
	dx_clinical_seizure	categorical	91.5	–	2	0 (92%), 1 (8%)		
	dx_stroke	categorical	91.5	–	2	0 (96%), 1 (4%)		
	dx_epilepsy	categorical	91.5	–	2	0 (97%), 1 (3%)		
	dx_parkinsons	categorical	91.5	–	2	0 (100%), 1 (0%)		
	dx_alzheimers	categorical	91.5	–	2	0 (100%), 1 (0%)		
	dx_tbi	categorical	91.5	–	2	0 (99%), 1 (1%)		
	dx_schizophrenia	categorical	91.5	–	2	0 (100%), 1 (0%)		
	dx_dementia	categorical	91.5	–	2	0 (100%), 1 (0%)		
	dx_substance	categorical	91.5	–	2	0 (96%), 1 (4%)		
clinical	**dx_anxiety**	categorical	91.5	–	2	0 (100%), 1 (0%)		
	dx_depression	categorical	91.5	–	2	0 (100%), 1 (0%)		
	dx_bipolar	categorical	91.5	–	2	0 (99%), 1 (1%)		
	dx_adhd	categorical	91.5	–	2	0 (100%), 1 (0%)		
	dx_ms	categorical	91.5	–	2	0 (100%), 1 (0%)		
	med_anesthetic	categorical	91.5	–	2	0 (94%), 1 (6%)		
	med_benzodiazepine	categorical	91.5	–	2	0 (78%), 1 (22%)		
	med_barbiturate	categorical	91.5	–	2	0 (96%), 1 (4%)		
	med_opioid	categorical	91.5	–	2	0 (87%), 1 (13%)		
	med_antiseizure	categorical	91.5	–	2	1 (50%), 0 (50%)		
	med_antipsychotic	categorical	91.5	–	2	0 (96%), 1 (4%)		
	f_firda	categorical	91.5	–	2	0 (98%), 1 (2%)		
	f_oirda	categorical	91.5	–	2	0 (100%), 1 (0%)		
report_findings	**f_tirda**	categorical	91.5	–	2	0 (100%), 1 (0%)		
	f_gpd	categorical	91.5	–	2	0 (100%), 1 (0%)		
	f_lpd	categorical	91.5	–	2	0 (99%), 1 (1%)		
qeeg	**qeeg_rel_delta**	quantitative	100.0	100.0	–		0.102 – 0.926	Delta (1-4 Hz) power / total 1-30 Hz power; high = slowing
	qeeg_rel_theta	quantitative	100.0	100.0	–		0.0362 – 0.425	Theta (4-8 Hz) power / total 1-30 Hz power; high = slowing / drowsiness

group	feature	type	TUEG %	TUAB %	n_categories	categories	active_range	meaning
	qeeg_rel_alpha	quantitative	100.0	100.0	–		0.0112 – 0.556	Alpha (8-13 Hz) power / total 1-30 Hz power; the normal awake rhythm
	qeeg_rel_beta	quantitative	100.0	100.0	–		0.00665 – 0.555	Beta (13-30 Hz) power / total 1-30 Hz power; fast activity
	qeeg_log_total_power	quantitative	100.0	100.0	–		0.336 – 4.22	log10 of the 1-30 Hz power [μV²]; low = low-voltage EEG
	qeeg_theta_alpha_ratio	quantitative	100.0	100.0	–		-0.775 – 0.949	log10(relative theta / relative alpha); rises as the background slows
	qeeg_slowing_ratio	quantitative	100.0	100.0	–		-0.589 – 1.69	log10((delta + theta) / (alpha + beta)); diffuse-slowing index
	qeeg_pdr_freq	quantitative	100.0	100.0	–		5 – 14	Posterior dominant rhythm: frequency of the largest 5-14 Hz peak above the 1/f fit over O1/O2/P3/P4 [Hz]; < 8 Hz in adults = slow
	qeeg_pdr_prominence	quantitative	100.0	100.0	–		-0.0166 – 1.54	Height of that peak above the 1/f fit [log10 power]; ~0 = no distinct PDR
	qeeg_aperiodic_slope	quantitative	100.0	100.0	–		-3.33 – -0.0275	Slope of log power vs log frequency, 2-30 Hz, posterior channels (the 1/f exponent); more negative = steeper
	qeeg_max_asymmetry	quantitative	100.0	100.0	–		0.0547 – 1.45	Largest |log10(left / right)| total power over 8 homologous pairs; hemispheric asymmetry
	qeeg_max_delta_asymmetry	quantitative	100.0	100.0	–		0.0598 – 1.47	Same for delta power only; focal (one-sided) slowing
	qeeg_median_kurtosis	quantitative	100.0	100.0	–		0.0823 – 4151	Median over channels of the excess kurtosis of the signal; spikes, sharp waves (and eye blinks) make it heavy-tailed

Table 8.3 shows what the larger cohort adds and loses. The clinical sheet covers 91.5% of recordings. TUAB has no counterpart at all for its 21 diagnosis and drug-class flags and 5 report findings. The header features (setup, channel inventory, history) are complete. The signal features are complete except for 2 recordings whose electrodes all carry the same trace (Section 11). The EKG lead is present in every header, but its signal is not in the npy, so Part I's heart-rate features have no TUEG column.

9. Demographics and history

The abnormal rate of a group G of labelled recordings, with $y_i = 1$ for abnormal, is

$$r(G) = \frac{1}{|G|} \sum_{i \in G} y_i. \tag{9.1}$$

Table 9.1. Age by label

Age (years) of the cohort recordings per label (all splits).

label3	count	mean	std	min	25%	50%	75%	max
normal	11604.0	48.3	17.1	18.0	34.0	48.0	61.0	89.0
abnormal	15515.0	55.5	17.2	11.0	44.0	57.0	68.0	89.0
unlabelled	2190.0	51.8	16.9	18.0	40.0	53.0	64.0	89.0

Figure 9.1. Age, sex and the abnormal rate

(a) Age histogram (5-year bins, density) of normal, abnormal and unlabelled recordings. (b) Abnormal rate, Eq. (9.1), per age group in the TUEG cohort (bars, with n), and in TUAB (Part I, black markers). Labelled recordings of all splits: n = 27,119 TUEG and 2,993 TUAB.

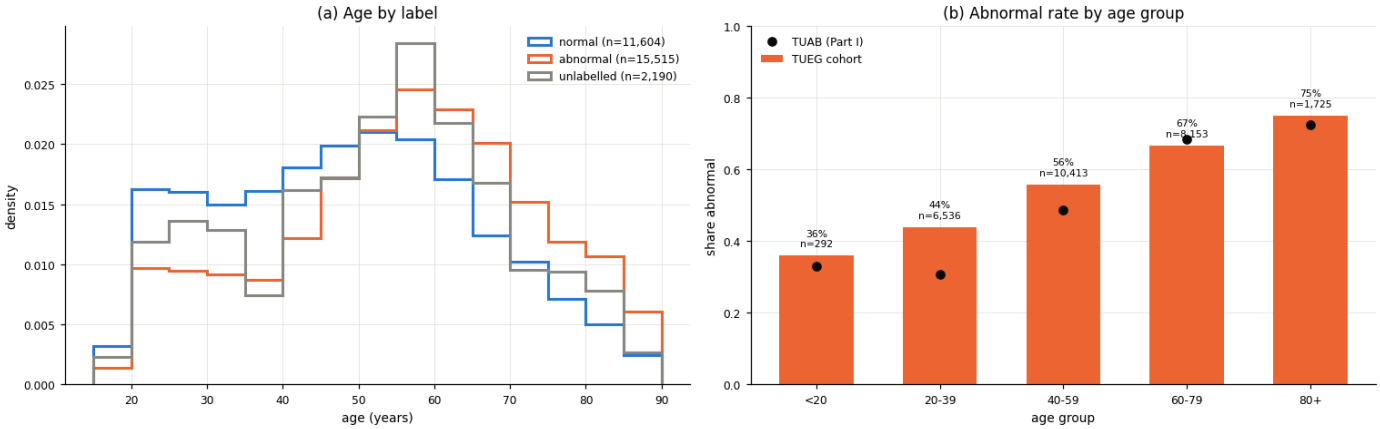


Table 9.2. Abnormal rate by sex, session number and recording year

Labelled recordings: count and abnormal rate, Eq. (9.1), per value. The session number says *which* TUH EEG session of the patient the recording is from (date order, whole TUH archive), not how many sessions the patient has; see Section 2. Years 1899 and 2000 are the placeholder dates noted in Part I.

variable	value	recordings	abnormal rate
sex	F	13,942	54%
	M	13,123	61%
session number	1	14,480	55%
	2	5,193	58%
	3	2,736	64%
	4-5	2,605	62%
	6-10	1,666	54%
	>10	439	64%
year	1899	155	47%
	2000	634	56%
	2002	23	4%
	2003	1,165	9%
	2004	1,111	59%
	2005	133	60%
	2006	1,456	58%
	2007	1,051	65%
	2008	1,202	68%
	2009	1,400	61%
	2010	1,539	63%
	2011	1,733	62%
	2012	2,754	56%
	2013	2,327	59%
	2014	3,438	54%
	2015	3,151	61%
	2016	3,783	60%
	2017	64	42%

Two TUAB demographics findings carry over and one does not (Figure 9.1, Table 9.2). Abnormal patients are again older, and the abnormal rate rises with age. In the cohort, though, the rate is higher in every age group, because TUEG is mostly abnormal: the 2017 TUAB release was balanced on purpose. Unlike TUAB, the session number is only weakly informative here: 55% of first sessions are abnormal against 54–64% later. TUAB's normal recordings were selected mostly from first sessions, which made the session number look like a strong proxy there. The abnormal rate is flat over 2004–2016 but drops to 4–9% in **2002–2003**, so the early recordings come from a different referral population or reporting standard (Section 10).

10. Acquisition setup: a shortcut for the label *and* for age

A setup variable is a shortcut when its value predicts the target without any brain activity. For normal/abnormal that is an abnormal rate, Eq. (9.1), that differs between its values. For brain age it is a mean age that differs between them. For a categorical variable with levels $k = 1..K$, n_k recordings and mean age $\bar{a}_k$ per level, the share of the age variance it explains (one-way η^2) and the spread of the abnormal rate are

$$\eta^2_{\text{age}} = \frac{\sum_k n_k \, (\bar{a}_k - \bar{a})^2}{\sum_i (a_i - \bar{a})^2}, \tag{10.1}$$

$$\Delta r = \max_{k:\, n_k \geq 100} r(G_k) - \min_{k:\, n_k \geq 100} r(G_k), \tag{10.2}$$

where a_i is the age (years) of recording i, $\bar{a}$ the cohort mean and G_k the labelled recordings at level k. Levels with fewer than 100 recordings are left out of Eq. (10.2), so tiny groups do not set the range.

Table 10.1. Shortcut strength of each setup variable

`eta2_age` is Eq. (10.1) over all recordings, `delta_abnormal_rate` is Eq. (10.2), and `levels` counts the values of the variable. Recording year and session number are binned as in Table 9.2, and the native sampling rate is taken as categorical.

variable	levels	eta2_age	delta_abnormal_rate
recording year	19	0.011	59%
amplifier family	6	0.005	58%
session number	6	0.004	10%
montage	3	0.004	3%
native sampling rate	5	0.004	56%
trend channels	2	0.004	4%
T1/T2 missing	2	0.003	7%
amplifier range (uV)	3	0.003	55%
sphenoidal electrodes	2	0.002	10%
EMG channel	2	0.001	1%
photic channel	2	0.000	2%
EOG (LOC/ROC)	2	0.000	2%

Table 10.2. Amplifier families

Recordings, abnormal rate (Eq. (9.1)), mean age (years), median recording year and share in the validation split per amplifier family. The family is the EDF `lrci_machine` tag (`unknown` = anonymised or unset tag).

equipment_family	recordings	abnormal_rate	mean_age	median_year	sfreq_mode	val_share
unknown	16,774	58%	51.7	2014	250	13%
Intution/Ultrasom	5,750	63%	54.5	2008	250	11%
Alliance	3,035	41%	53.2	2004	250	10%
Compact_32ch	2,170	64%	52.4	2015	256	10%
Compact_64ch	1,406	58%	50.0	2015	256	12%
Bravo	174	7%	47.5	2003	400	13%

Figure 10.1. Recording era and amplifier

(a) Recordings per year by amplifier family (stacked; placeholder years 1899/2000 left out). (b) Mean age (years) per recording year, normal and abnormal recordings separately. TUEG cohort, all splits.

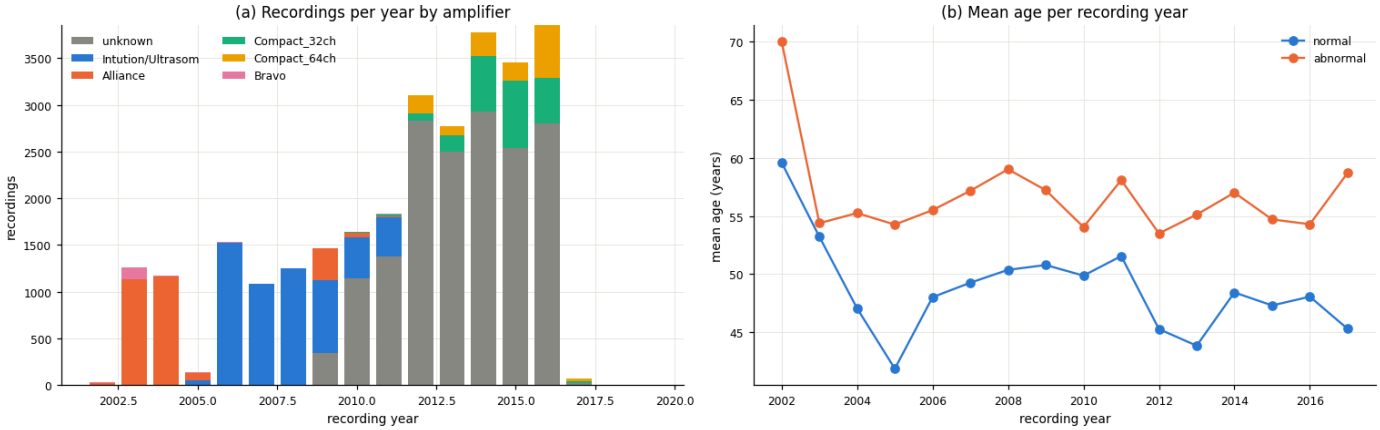


Table 10.1, Table 10.2 and Figure 10.1 show that the setup matters for the label but not for age.

For the label, the recording year and the amplifier are the strongest shortcuts: the abnormal rate varies by 58–59 percentage points across their values. One amplifier family (Bravo: 400 Hz sampling rate and a ±29,483 µV range, 2002–2003 only, 174 recordings) is **7% abnormal**, and all recordings from 2002–2003 are only 4–9% abnormal, against 54–68% in every later year. Its sampling rate and amplitude range leave a fingerprint even after resampling, so a model can learn "this amplifier = normal".

For age, no setup variable explains more than 1.1% of the age variance (the recording year, Eq. (10.1)). Even the shift from linked-ears (`02_tcp_le`, median year 2006) to average reference (`01_tcp_ar`, median year 2014) comes with only a 2.4-year difference in mean age. The cohort therefore offers no strong setup shortcut for brain age; Table 14.3 confirms this from the model side.

11. Signal quality of the training data

These are the Part I quality measures (`labram/data/tuh_meta_features.py::quality_features`), here computed on the processed npy the model actually reads. With E 10-s epochs and the recorded scalp channels c (zero-filled channels are excluded), after a 0.5–45 Hz band-pass:

$$b = \frac{1}{E}\sum_{e=1}^{E} 1\Big[\exists c:\ \sigma_{ec} < 0.5\,\mu\mathrm{V}\ \vee\ \mathrm{ptp}_{ec} > 500\,\mu\mathrm{V}\Big], \tag{11.1}$$

where σ_{ec} and ptp_{ec} are the standard deviation and peak-to-peak range of channel c in epoch e. A channel is *flat* when $\sigma_{ec} < 0.5\,\mu$V in more than half of the epochs. It is *noisy* when its log standard deviation over the whole recording is a robust outlier among the recording's channels:

$$\frac{|\log\sigma_c - \mathrm{med}_{c'}\log\sigma_{c'}|}{1.4826\ \mathrm{MAD}_{c'}(\log\sigma_{c'})} > 5. \tag{11.2}$$

The quality tier is the Part I rule without its clipping criterion (clipping cannot be measured after preprocessing): *poor* if ≥ 2 flat channels or $b > 0.3$; else *fair* if ≥ 1 flat or noisy channel or $b > 0.1$; else *good*.

Figure 11.1. Artifact load per recording

(a) Bad-epoch fraction b, Eq. (11.1), per recording by label (log count axis). (b) Flat + noisy electrodes per recording, Eq. (11.2), by label (log axis). TUEG cohort, all splits.

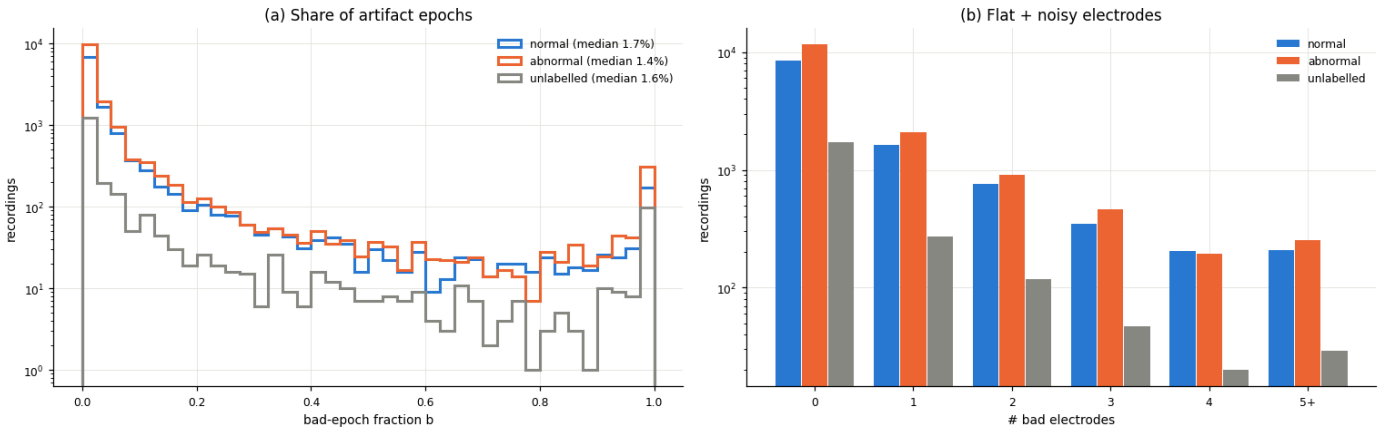


Table 11.1. Quality tiers by split and label

Recordings per quality tier (rule above), with the share of the row in parentheses, for the TUEG cohort. The last row is TUAB from Part I. It is not directly comparable, because there the tier was measured on the raw EDF and also counts clipping.

split	label3	quality_tier	good	fair	poor
test	abnormal		88 (70%)	33 (26%)	4 (3%)
	normal		105 (70%)	42 (28%)	2 (1%)
train	abnormal		9221 (67%)	3485 (25%)	1008 (7%)
	normal		6494 (65%)	2807 (28%)	766 (8%)
	unlabelled		1198 (65%)	383 (21%)	251 (14%)
val	abnormal		1144 (68%)	417 (25%)	115 (7%)
	normal		891 (64%)	394 (28%)	103 (7%)
	unlabelled		229 (64%)	80 (22%)	49 (14%)
TUAB (Part I)	all		2116 (71%)	791 (26%)	86 (3%)

`recordings whose scalp channels are all identical: 2 (splits: {'train': 2})`

Table 11.1 and Figure 11.1 differ from TUAB in two ways.

- **More artifact.** 7–8% of the labelled train and val recordings are *poor*, against 3% in TUAB, even though clipping is no longer counted here. TUEG keeps long ICU and ambulatory recordings that the curated TUAB set does not have (median duration 20 min, but up to 24 h; the first 60 min are analysed).
- **Not label-dependent.** Poor quality hits normal and abnormal recordings alike (8% vs 7% in train). The *unlabelled* recordings, those not in the clinical sheet, are the worst: 14% poor.

Two train recordings are corrupt: all 19 scalp electrodes carry the identical trace. No electrode is flat, so the quality measures above do not catch them. Only the common average gives them away, because it leaves no signal at all (their qEEG features are NaN). They should be dropped from the brain-age training split.

12. Clinical context from the NEDC sheet

None of this exists for TUAB. For a flag d (diagnosis or drug class) over the labelled recordings of sessions in the sheet, with D the recordings where d is true and $\bar{D}$ the rest:

$$\mathrm{prev}(d) = \frac{|D|}{|D| + |\bar{D}|}, \qquad \Delta_{\mathrm{age}}(d) = \bar{a}(D) - \bar{a}(\bar{D}), \tag{12.1}$$

with $\bar{a}$ the mean age in years. The abnormal rates $r(D)$ and $r(\bar{D})$ follow Eq. (9.1). Drug classes come from token matching of the medication list (`dataset_maker/tuh_clinical.py::DRUG_CLASSES`). *Sedated* = anaesthetic, benzodiazepine, barbiturate or opioid. Diagnoses are the sheet's Y/N columns.

Table 12.1. Diagnoses and medication classes

One row per flag with at least 30 positive recordings: recordings and patients with the flag, `prev` and `delta_age`, Eq. (12.1), and the abnormal rate with and without it. Sorted by recordings.

flag	recordings	patients	prev	delta_age	abnormal_with	abnormal_without
med_antiseizure	13,533	6,293	50.5%	-3.4	62%	51%
sedated	8,816	4,282	32.9%	+0.8	62%	54%
med_benzodiazepine	5,889	2,827	22.0%	+0.0	61%	56%
med_opioid	3,534	1,924	13.2%	+2.5	56%	57%
any_neuro_dx	3,392	1,583	12.6%	-0.6	65%	56%
dx_clinical_seizure	2,067	999	7.7%	-4.6	58%	57%
med_anesthetic	1,507	466	5.6%	+5.4	52%	57%
med_antipsychotic	1,177	761	4.4%	-1.3	63%	57%
dx_substance	1,166	677	4.3%	-4.8	71%	56%
dx_stroke	1,049	482	3.9%	+11.1	82%	56%
med_barbiturate	961	450	3.6%	-1.5	64%	57%
dx_epilepsy	822	280	3.1%	-2.0	61%	57%
dx_electrographic_seizure	358	53	1.3%	+5.3	84%	57%
dx_tbi	348	179	1.3%	-5.7	69%	57%
dx_bipolar	314	229	1.2%	-8.1	59%	57%
dx_depression	126	74	0.5%	-6.4	44%	57%
dx_anxiety	121	70	0.5%	-10.0	48%	57%
dx_schizophrenia	90	40	0.3%	-5.9	52%	57%
dx_dementia	78	43	0.3%	+22.7	72%	57%
dx_ms	60	37	0.2%	-5.8	47%	57%
dx_adhd	35	18	0.1%	-21.4	43%	57%

Table 12.2. TUAB folder label vs NEDC report

The cohort's TUAB recordings that are also in the clinical sheet: the TUAB folder label (rows) against the sheet's `abnormal due to` field being non-empty (columns). The agreement is the diagonal share.

abnormal_report abnormal	report abnormal	report normal
TUAB abnormal	903	79
TUAB normal	26	1316

`agreement: 95.5% of 2,324 recordings`

Figure 12.1. Who are the patients?

(a) Prevalence, Eq. (12.1), of the most frequent diagnoses per age group (labelled recordings in the sheet). (b) Prevalence of each medication class in normal vs abnormal recordings. TUEG cohort, all splits; n = 26,819 labelled recordings in the sheet.

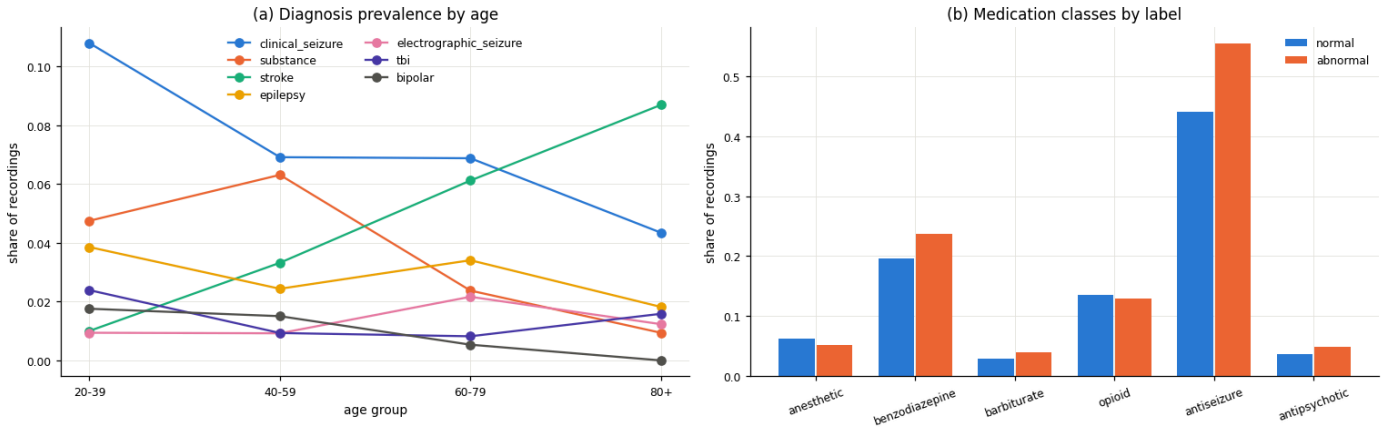


Table 12.1 and Figure 12.1 describe a **clinical epilepsy and ICU population**, not a healthy one. Half of all recordings list an antiseizure drug, a fifth a benzodiazepine, and a third are

sedated by the Eq. (12.1) definition. Stroke patients are 11 years older than the rest, and dementia patients 23 years older. Electrographic seizures, stroke and dementia are the most often read abnormal (84%, 82% and 72%). Psychiatric diagnoses (anxiety, depression, ADHD) sit below the 57% base rate. The TUAB folder label and the report agree on 95.5% of the 2,324 shared recordings (Table 12.2). The report-derived label used for the non-TUAB recordings is therefore a fair stand-in.

For **brain age** this means two things. A "normal" EEG here is a clinically normal read in a patient who still had a reason to be referred. And age is confounded with age-related disease and with medication: among report-normal recordings, anaesthetic and opioid use correlate with age (Figure 14.1(a)) and change the spectrum (Table 13.5). The normative subsets of `tuh_clinical.normative_mask` exist for this reason.

13. Why is a case abnormal? The report's reason vs the qEEG signatures

Part I could only *guess* why a TUAB recording is abnormal: from qEEG signatures, a qEEG feature beyond the 95th (5th) percentile of age-matched normal recordings. In the cohort, the sheet's free-text `abnormal due to` field states the reason, so the guesses can be checked.

Report reasons. `labram/eval/meta_analysis.py::report_reason_flags` maps the text onto 8 categories by keyword rules (`REPORT_REASONS`). They are multi-label: "diffuse slowing and left temporal sharp waves" is both *diffuse slowing* and *epileptiform*. Keyword rules are approximate: "focal background slowing" also counts as diffuse.

Signatures. For signature s on qEEG feature x with direction "high" (Part I, `SIGNATURES`), recording i in age group $g(i)$ is flagged when

$$f_{is} = 1\left[x_i > Q_{0.95}\big(\{x_j : j \in \mathcal{R},\ g(j) = g(i)\}\big)\right], \tag{13.1}$$

and "low" signatures use $x_i < Q_{0.05}(\cdot)$. Here $\mathcal{R}$ is the reference set: labelled-normal recordings of the cohort's **train** split. An age group with fewer than 20 reference recordings uses the pooled reference. For a reason c and its matching signature $s(c)$ (`REASON_TO_SIGNATURE`), the **sensitivity** and **lift** are

$$\mathrm{sens}(c) = \frac{\sum_{i \in A_c} f_{i\,s(c)}}{|A_c|}, \qquad \mathrm{lift}(c) = \frac{\mathrm{sens}(c)}{\frac{1}{|N|}\sum_{i \in N} f_{i\,s(c)}}, \tag{13.2}$$

where A_c are the abnormal recordings whose report gives reason c, and N the normal recordings. By Eq. (13.1), $\frac{1}{|N|}\sum_{i \in N} f_{is} \approx 5\%$.

Table 13.1. Reasons given by the abnormal reports

Abnormal recordings in the sheet with a non-empty reason (a recording can be in several rows). `share` is the share of those recordings; `mean age` is in years.

	recordings	share	mean age
diffuse_slowing	9,351	61.4%	57.2
focal_slowing	6,768	44.4%	56.4
epileptiform	5,001	32.8%	54.8
suppression	2,971	19.5%	55.3
pdr_abnormal	1,212	8.0%	55.6
excess_beta	1,031	6.8%	49.0
asymmetry	1,009	6.6%	57.9
seizure	621	4.1%	55.1
reason_unmatched	357	2.3%	53.5

`15,227 abnormal recordings with a reason; 56% have more than one category`

Table 13.2. Signature prevalence, TUEG cohort vs TUAB

Share of normal and abnormal recordings flagged by each signature, Eq. (13.1). TUEG cohort (all splits; reference = cohort train normals) next to the Part I TUAB numbers.

	normal	abnormal	lift	TUAB normal	TUAB abnormal	TUAB lift
diffuse_slowing	4.8%	14.2%	x3.0	5.1%	49.9%	x9.8
asymmetry	4.9%	9.2%	x1.9	5.5%	21.9%	x4.0
focal_slowing	4.9%	9.0%	x1.8	5.3%	29.7%	x5.6
slow_pdr	4.3%	7.7%	x1.8	3.8%	27.9%	x7.3
absent_pdr	5.1%	6.6%	x1.3	4.8%	11.5%	x2.4
low_voltage	5.0%	5.0%	x1.0	5.4%	2.9%	x0.5
sharp_transients	5.1%	3.8%	x0.8	5.6%	3.7%	x0.7

`abnormal with >= 1 signature: 35%, normal: 24%`

Table 13.3. Do the signatures find the reason the report gives?

For each report reason c: abnormal recordings with that reason ($|A_c|$), the matching signature $s(c)$, its sensitivity and lift, Eq. (13.2), and how many of these recordings have *no* signature at all. Reasons with no whole-recording spectral proxy (seizure, excess beta) have no signature.

reason	recordings	signature	no signature	sensitivity	lift
diffuse_slowing	9,351	diffuse_slowing	63%	18%	x3.8
focal_slowing	6,768	focal_slowing	70%	10%	x2.1
pdr_abnormal	1,212	slow_pdr	64%	7%	x1.7
asymmetry	1,009	asymmetry	68%	7%	x1.5
suppression	2,971	low_voltage	48%	18%	x3.6
epileptiform	5,001	sharp_transients	65%	2%	x0.5
seizure	621	–	58%	–	–
excess_beta	1,031	–	68%	–	–

Figure 13.1. Report reasons against qEEG signatures

(a) Share of abnormal recordings with report reason c (rows) flagged by each signature (columns), Eq. (13.1); the bottom row is the normal recordings for reference. (b) Number of signatures per recording by label (share of the class). TUEG cohort, all splits.

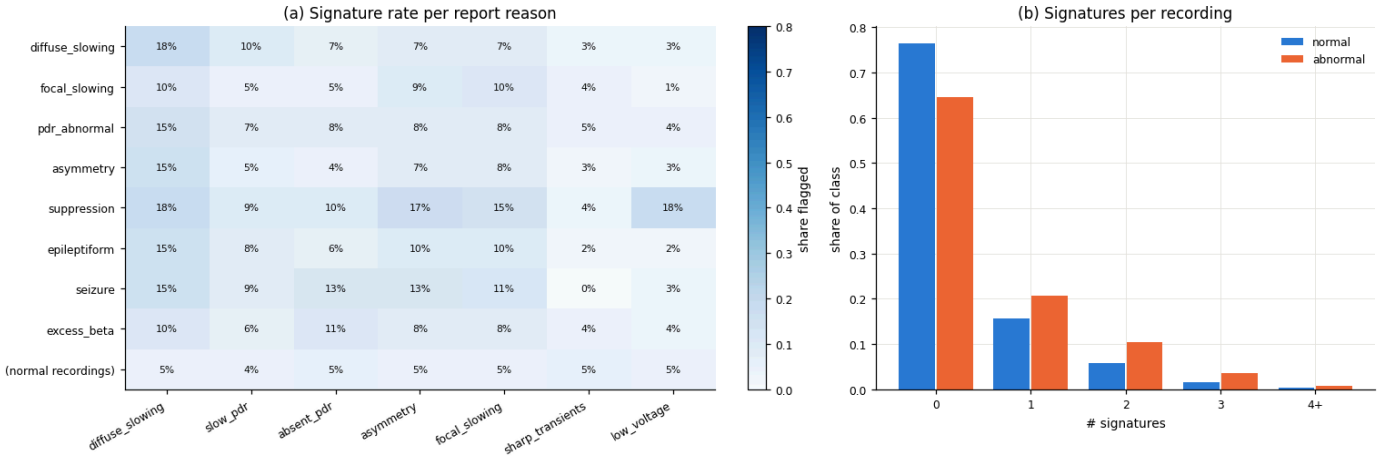


Table 13.4. Signatures by diagnosis

Lift of each signature in recordings with a diagnosis vs without it, `labram/eval/meta_analysis.py::signature_by_condition`. Abnormal *and* normal labelled recordings in the sheet, diagnoses with ≥ 30 positives. The top three signatures per diagnosis by lift are shown.

condition	signature	n_positive	prev_with	prev_without	lift
dx_adhd	absent_pdr	35	17%	6%	x2.9
	slow_pdr	35	17%	6%	x2.8
	focal_slowing	35	14%	7%	x2.0
dx_anxiety	sharp_transients	121	7%	4%	x1.5
	focal_slowing	121	6%	7%	x0.8
	slow_pdr	121	5%	6%	x0.8
dx_bipolar	sharp_transients	314	5%	4%	x1.2
	low_voltage	314	4%	5%	x0.8
	absent_pdr	314	5%	6%	x0.8
dx_clinical_seizure	sharp_transients	2,067	7%	4%	x1.7
	absent_pdr	2,067	8%	6%	x1.3
	diffuse_slowing	2,067	10%	10%	x1.0
dx_dementia	slow_pdr	78	12%	6%	x1.9
	sharp_transients	78	8%	4%	x1.7
	diffuse_slowing	78	13%	10%	x1.3
dx_depression	sharp_transients	126	10%	4%	x2.2
	absent_pdr	126	9%	6%	x1.5
	slow_pdr	126	6%	6%	x0.9
dx_electrographic_seizure	absent_pdr	358	11%	6%	x1.9
	focal_slowing	358	11%	7%	x1.6
	diffuse_slowing	358	14%	10%	x1.3
dx_epilepsy	absent_pdr	822	9%	6%	x1.6
	focal_slowing	822	7%	7%	x1.0
	sharp_transients	822	5%	4%	x1.0
dx_ms	absent_pdr	60	8%	6%	x1.4
	sharp_transients	60	5%	4%	x1.1
	diffuse_slowing	60	10%	10%	x1.0
dx_schizophrenia	diffuse_slowing	90	19%	10%	x1.9
	absent_pdr	90	9%	6%	x1.5
	sharp_transients	90	6%	4%	x1.3
dx_stroke	diffuse_slowing	1,049	17%	10%	x1.7
	slow_pdr	1,049	8%	6%	x1.3
	focal_slowing	1,049	9%	7%	x1.2
dx_substance	diffuse_slowing	1,166	16%	10%	x1.6
	low_voltage	1,166	6%	5%	x1.3
	asymmetry	1,166	8%	7%	x1.2
dx_tbi	diffuse_slowing	348	25%	10%	x2.5
	asymmetry	348	17%	7%	x2.3
	focal_slowing	348	15%	7%	x2.1

Table 13.5. Drug effects on the qEEG of normal recordings

Report-normal recordings only, so pathology does not drive it. For each medication class: recordings on it, and the median relative beta power, slowing ratio and PDR frequency (Hz) with (+) and without (−) the class. `AUC beta` is the univariate ROC-AUC of relative beta power for the class against the rest, oriented so that > 0.5 means more beta on the drug.

drug class	recordings	beta +	beta -	slowing +	slowing -	PDR + (Hz)	PDR - (Hz)	AUC beta
med_anesthetic	718	0.065	0.151	0.566	0.206	9.500	10.000	0.27
med_benzodiazepine	2,271	0.132	0.148	0.360	0.200	10.000	9.500	0.47
med_barbiturate	343	0.130	0.146	0.444	0.222	10.000	10.000	0.45
med_opioid	1,561	0.100	0.152	0.432	0.199	9.500	10.000	0.37
med_antiseizure	5,073	0.145	0.147	0.267	0.202	9.500	10.000	0.49
med_antipsychotic	433	0.149	0.146	0.230	0.228	9.500	10.000	0.51
sedated	3,347	0.125	0.152	0.377	0.175	10.000	10.000	0.44

Section 13 checks Part I's computed "why abnormal" signatures against what the reports actually say, and they come out **much weaker than TUAB suggested**.

- **What the reports say (Table 13.1).** Slowing dominates: diffuse slowing in 61% of the 15,227 abnormal reports with a reason and focal slowing in 44%. Epileptiform findings (spikes, sharp waves, periodic discharges) are third at 33%, then suppression or attenuation at 20%. 56% of reports give more than one reason.
- **The signatures barely separate the classes here (Table 13.2).** 35% of abnormal and 24% of normal recordings carry at least one signature, against 72% vs 27% in TUAB. Diffuse slowing, the best signature, has a lift of ×3.0 instead of ×9.8.
- **Against the report's own reason (Table 13.3, Figure 13.1(a)).** The matching signature fires on 18% of the recordings whose report names diffuse slowing (×3.8 the rate in normals). For a report of suppression, the low-voltage signature fires on 18% (×3.6). For focal slowing, slow PDR and asymmetry it fires on only 7–10% (×1.5–2.1). Epileptiform findings are invisible to the kurtosis proxy (2%, ×0.5), as Part I suspected. Whatever the reason, 58–70% of these abnormal recordings carry *no* signature at all (48% for suppression).

Several things plausibly weaken the signatures compared with TUAB. The reference normals are clinical "normal reads" rather than TUAB's curated normals. Whole recordings of up to 60 min mix wakefulness, drowsiness and sleep, and sleep alone looks like slowing. And a report's reason often describes something brief or mild that a recording-wide median spectrum averages away. TUAB's ×9.8 is therefore an optimistic estimate of how much 13 spectral summaries explain.

Diagnoses (Table 13.4) move the signatures only modestly. Traumatic brain injury is the clearest case (diffuse slowing ×2.5, asymmetry ×2.3, focal slowing ×2.1); stroke raises diffuse slowing (×1.7) and dementia a slow PDR (×1.9). Small groups (ADHD, n = 35) produce noisy lifts. **Medication (Table 13.5)** leaves a large fingerprint on report-normal EEGs. Anaesthetics nearly triple the slowing ratio (0.57 vs 0.21) and halve relative beta, and benzodiazepines, barbiturates and opioids also raise slowing. Relative beta is *lower* on these drugs (AUC 0.27–

0.47). These are relative powers, so extra slow activity can hide an absolute increase in fast activity. For a brain-age model, sedation therefore looks like an "older", slower brain.

14. Can the meta features be used? Normal/abnormal and age

Metadata-only gradient-boosted trees (`metadata_baseline` , `metadata_age_baseline` in `labram/eval/meta_analysis.py`) are trained on the cohort's train split and scored on the subject-disjoint **val** split and on the **test** split (TUAB eval), one feature group at a time. The metrics on the n scored recordings, with y_i the label, p_i the predicted probability, a_i the age and $\hat{a}_i$ the predicted age (years):

$$\mathrm{AUROC} = \Pr\left(p_i > p_j \mid y_i = 1,\ y_j = 0\right), \tag{14.1}$$

$$\mathrm{BalAcc} = \tfrac{1}{2}\left(\frac{\sum_i 1[p_i \geq 0.5,\ y_i = 1]}{\sum_i y_i} + \frac{\sum_i 1[p_i < 0.5,\ y_i = 0]}{\sum_i (1 - y_i)}\right), \tag{14.2}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_i |\hat{a}_i - a_i|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i (\hat{a}_i - a_i)^2}, \tag{14.3}$$

$$R^2 = 1 - \frac{\sum_i (\hat{a}_i - a_i)^2}{\sum_i (a_i - \bar{a})^2}, \qquad \mathrm{MAE}_0 = \frac{1}{n}\sum_i |\bar{a}_{\mathrm{train}} - a_i|. \tag{14.4}$$

MAE_0 is the error of always predicting the mean training age: the floor any age model has to beat. The `report findings` group (FIRDA, GPD, ...) is the report's own EEG reading. It is listed only to show how much it leaks; a usable model must not use it.

Table 14.1. Label information per meta feature

The 15 features with the highest univariate ROC-AUC (Eq. (14.1) of the feature itself, oriented so 0.5 = none, `labram/eval/meta_analysis.py::univariate_auc`) on the labelled train recordings. `coverage` is Eq. (8.1) within those recordings.

feature	group	auc	direction	coverage	median_normal	median_abnormal
qeeg_theta_alpha_ratio	qeeg	0.690	higher=abnormal	100%	-0.023	0.234
qeeg_slowing_ratio	qeeg	0.679	higher=abnormal	100%	0.234	0.606
qeeg_rel_alpha	qeeg	0.668	lower=abnormal	100%	0.174	0.098
qeeg_aperiodic_slope	qeeg	0.661	lower=abnormal	100%	-1.487	-1.909
qeeg_rel_beta	qeeg	0.656	lower=abnormal	100%	0.145	0.084
qeeg_rel_delta	qeeg	0.644	higher=abnormal	100%	0.451	0.597
age	demographics	0.617	higher=abnormal	100%	49.000	57.000
qeeg_max_delta_asymmetry	qeeg	0.614	higher=abnormal	100%	0.183	0.239
qeeg_pdr_prominence	qeeg	0.607	lower=abnormal	100%	0.500	0.365
qeeg_pdr_freq	qeeg	0.607	lower=abnormal	100%	10.000	8.500
q_median_amp_uv	quality	0.604	higher=abnormal	100%	9.172	10.834
qeeg_log_total_power	qeeg	0.603	higher=abnormal	100%	1.859	2.028
qeeg_median_kurtosis	qeeg	0.594	lower=abnormal	100%	8.702	3.916
qeeg_max_asymmetry	qeeg	0.589	higher=abnormal	100%	0.176	0.215
med_antiseizure	clinical	0.557	higher=abnormal	99%	0.000	1.000

Table 14.2. Metadata-only normal/abnormal classifier

AUROC (Eq. (14.1)) and balanced accuracy (Eq. (14.2)) per feature set, trained on the labelled train recordings (n = 23,781) and scored on the labelled val (n = 3,064) and test (TUAB eval, n = 274) recordings. The TUAB column repeats Part I's result for the same feature set where one exists.

features	n_features	val AUROC	val BalAcc	test AUROC	test BalAcc	TUAB AUROC (Part I)
demographics (age, sex)	2	0.628	0.569	0.693	0.611	0.689
+ history (session no., year)	4	0.660	0.597	0.711	0.620	0.727
acquisition only	20	0.620	0.569	0.505	0.486	0.686
quality only	6	0.654	0.586	0.718	0.647	0.574
clinical context only (dx, drugs)	21	0.610	0.572	0.650	0.632	–
qEEG only	13	0.767	0.704	0.907	0.828	0.890
demographics + qEEG	15	0.769	0.706	0.915	0.854	0.901
demographics + qEEG + clinical	36	0.786	0.711	0.927	0.850	–
all except acquisition/history/findings	42	0.790	0.717	0.931	0.858	–
report findings only (leak)	5	0.534	0.500	0.536	0.500	–

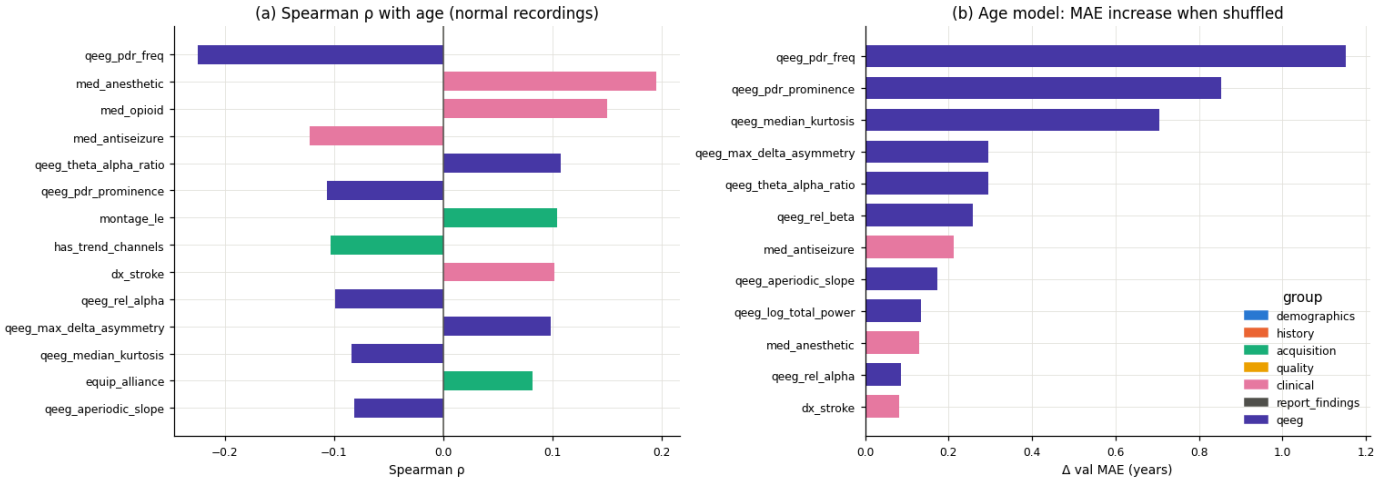
Table 14.3. Metadata-only age regression

MAE, RMSE (Eq. (14.3)), R^2 and MAE_0 (Eq. (14.4)), all in years, of gradient-boosted age regression per feature set (age itself excluded). Trained on all train recordings (n = 25,613) and scored on all val recordings and on the report-normal val recordings.

features	n_features	val MAE	val RMSE	val R2	val-normal MAE	MAE0 (val)
sex only	1	13.68	16.82	-0.00	13.91	13.68
history (session no., year)	3	13.59	16.73	0.01	13.67	13.68
acquisition only	20	13.23	16.41	0.05	13.25	13.68
acquisition + history	23	13.16	16.36	0.05	13.05	13.68
quality only	6	13.54	16.67	0.02	13.76	13.68
clinical context only (dx, drugs)	21	13.34	16.53	0.03	13.44	13.68
qEEG only	13	12.24	15.17	0.19	12.12	13.68
qEEG + sex	14	12.25	15.20	0.18	12.12	13.68
qEEG + sex + quality + clinical	41	11.96	14.82	0.22	11.74	13.68
everything (incl. setup, history, label)	69	11.78	14.59	0.25	11.39	13.68

Figure 14.1. What tracks age, and what the age model uses

(a) Spearman ρ between each meta feature and age within **report-normal** recordings, so pathology does not drive it; the 14 strongest. (b) Increase in val MAE (years) when a feature of the "qEEG + sex + quality + clinical" age model is shuffled (permutation importance, 5 repeats); the 12 strongest. TUEG cohort.



Normal/abnormal (Table 14.1, Table 14.2). On the TUEG val split the metadata-only model is much weaker than on TUAB:

- qEEG alone reaches AUROC 0.767 (val), and all patient-level groups together 0.790;
- the best single features are the theta/alpha and slowing ratios (univariate AUC ≈ 0.68–0.69).

On the **TUAB eval** test set the same TUEG-trained qEEG model scores 0.907, *above* the TUAB-trained 0.890 of Part I. So the cohort is harder, not the model worse: the TUEG abnormal class is broader and subtler (Section 13). The other groups:

- **Clinical context** adds a little (0.769 → 0.786 val when added to demographics + qEEG). It is session-level knowledge from the same referral as the label, so it is only usable where it is known at prediction time.
- **Report findings** (FIRDA, LPD, ...) mark abnormal recordings by definition but are too rare (≈ 2%) to matter: AUROC 0.53.
- **Acquisition features** reach 0.620 on val, through the year / amplifier shortcut of Section 10. They collapse to 0.505 on TUAB eval, which has no 2002–2003 recordings.

Age (Table 14.3, Figure 14.1). The meta features say little about age.

- The trivial predictor (mean training age) has MAE_0 = 13.7 years.
- Setup and history reach 13.2 (R^2 = 0.05), confirming Section 10: no setup shortcut.
- qEEG + sex reaches 12.2 (R^2 = 0.18), and everything together, including the label, 11.8 (R^2 = 0.25).

Among normal recordings, no feature correlates with age beyond $|\rho| = 0.23$. The strongest is the PDR frequency, slower with age ($\rho = -0.23$). Medication comes next: anaesthetic and opioid use rise with age ($\rho = +0.20, +0.15$). LaBraM's brain-age fine-tunes reach about 6.3–6.5 years on the TUAB part of val (`docs/data/tuh_cohort_eval_v1.csv`), half the error of these 13 summaries. Nearly all the age information is therefore in the raw EEG, not in coarse spectral summaries.

15. TUEG cohort: findings and recommendations

Table 15.1. TUAB vs TUEG cohort at a glance

The headline numbers of Part I and Part II side by side (computed from the tables above).

	TUAB (Part I)	TUEG cohort (Part II)
recordings / subjects	2,993 / 2,329	29,309 / 13,801
abnormal share (labelled)	49%	57%
mean age normal / abnormal	44.0 / 54.4	48.3 / 55.5
diagnoses / drugs / report reason	none	92% of recordings
abnormal with >= 1 signature	72%	35%
poor-quality tier	2.9%	7.8%
qEEG-only AUROC (eval)	0.890	0.907 (val 0.767)
qEEG + sex age MAE (years)	–	12.25 (MAE0 13.68)

Recommendations

1. **Drop the 2 corrupt train recordings** (identical trace on every electrode, Section 11). Add an "all channels identical" check to the cohort builder; the existing flat/noisy checks cannot see it.
2. **Stratify brain-age evaluation by medication and diagnosis**, not only by normal/abnormal. Report MAE and age bias for sedated vs non-sedated, and for stroke / dementia / TBI vs none, on the val split. Sedation slows report-normal EEGs strongly (Table 13.5) and is more common in older patients, so it can masquerade as brain ageing.
3. **Treat the 2002–2003 / Bravo recordings as a label shortcut.** They are 4–9% abnormal and carry a distinct amplifier fingerprint (Tables 9.2, 10.2). Check abnormal detection within amplifier families, or exclude them from label-supervised training. For age they are harmless (Table 10.1).
4. **Use the report reasons for error analysis, not the qEEG signatures.** The signatures miss most report-stated reasons in TUEG (Table 13.3). The keyword categories of `report_reason_flags` give a per-recording "why" for 15k abnormal recordings: score abnormal-detection recall per reason, especially for epileptiform-only reports.
5. **Do not feed setup, history or report findings to a model** (Table 14.2), for the same reason as in Part I. Clinical flags are legitimate *conditioning* inputs only where they are known at prediction time; otherwise use them for stratification.
6. **Do not use these 13 qEEG summaries as the brain-age bar.** They reach $R^2 = 0.18$ (Table 14.3), so beating them proves little. Compare LaBraM against the richer handcrafted feature set of `notebooks/eeg_feature_extraction_tuab.ipynb` instead.

7. **The *unlabelled* 7.5% (2,190 recordings)** have neither a TUAB label nor a sheet entry. They are fine for age, but any normal-only cohort must keep excluding them, as `normative_mask` does.